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(54) **OPTICAL SYSTEM FOR IMAGING PICKUP**

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(51) **Int. Cl.**

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G02B 13/18 (2006.01)

G02B 3/02 (2006.01)

(52) **U.S. Cl.**

USPC **359/764**; **359/714**

(58) **Field of Classification Search**

USPC **359/763**, **764**, **714**, **649**, **746**

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,480,105 B2 1/2009 Mori
7,502,181 B2 3/2009 Shinohara
2011/0310287 A1* 12/2011 Ohtsu 348/340

* cited by examiner

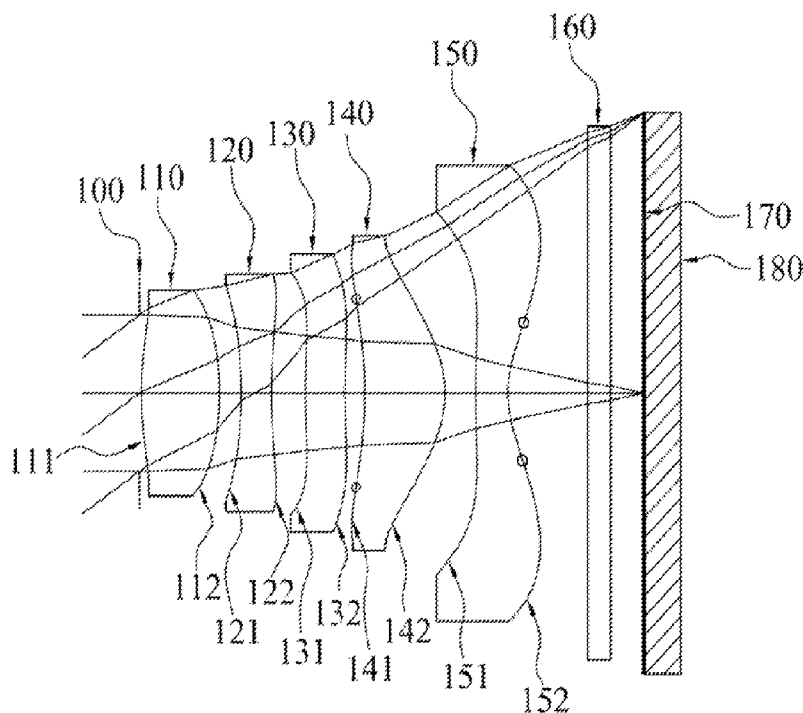
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(57) **ABSTRACT**

An optical system for imaging pickup includes, sequentially arranged from an object side to an image side: the first lens element with positive refractive power having a convex image-side surface, the second lens element with refractive power having a concave image-side surface, the third lens element with refractive power, the fourth lens element with refractive power having both object-side and image-side surfaces being aspheric, and the fifth lens element with negative refractive power having a concave image-side surface, and both object-side and image-side surfaces being aspheric. By such arrangements, the optical system for imaging pickup satisfies conditions related to shorten the total length and to reduce the sensitivity for use in compact cameras and mobile phones with camera functionalities.

22 Claims, 14 Drawing Sheets



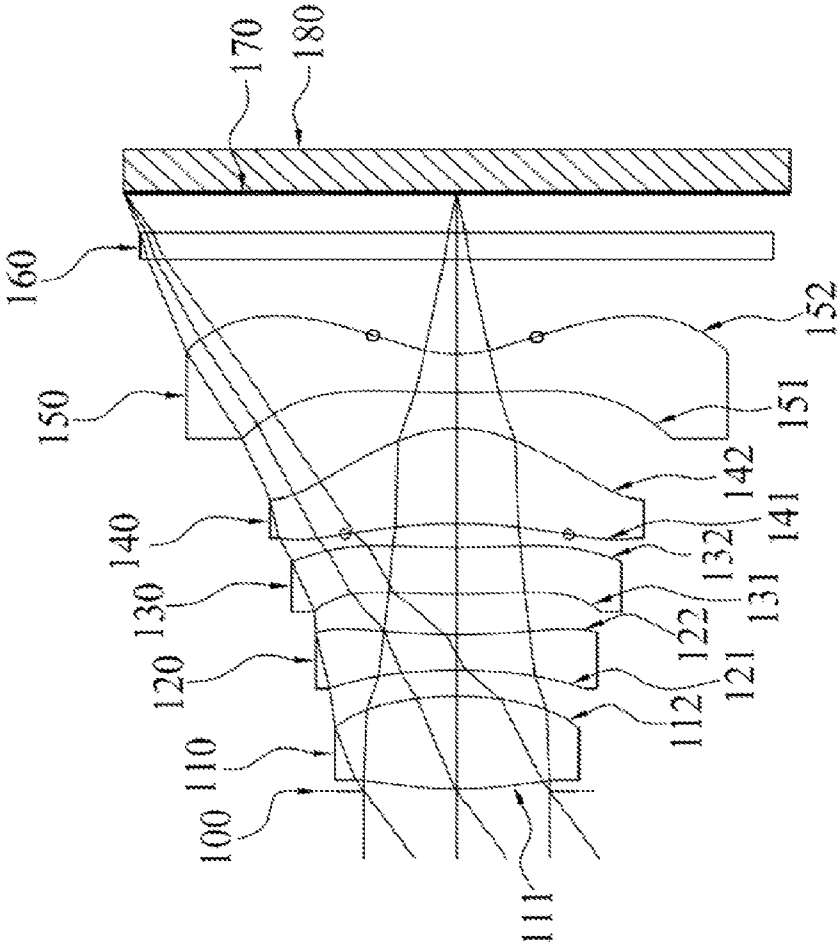


FIG. 1A

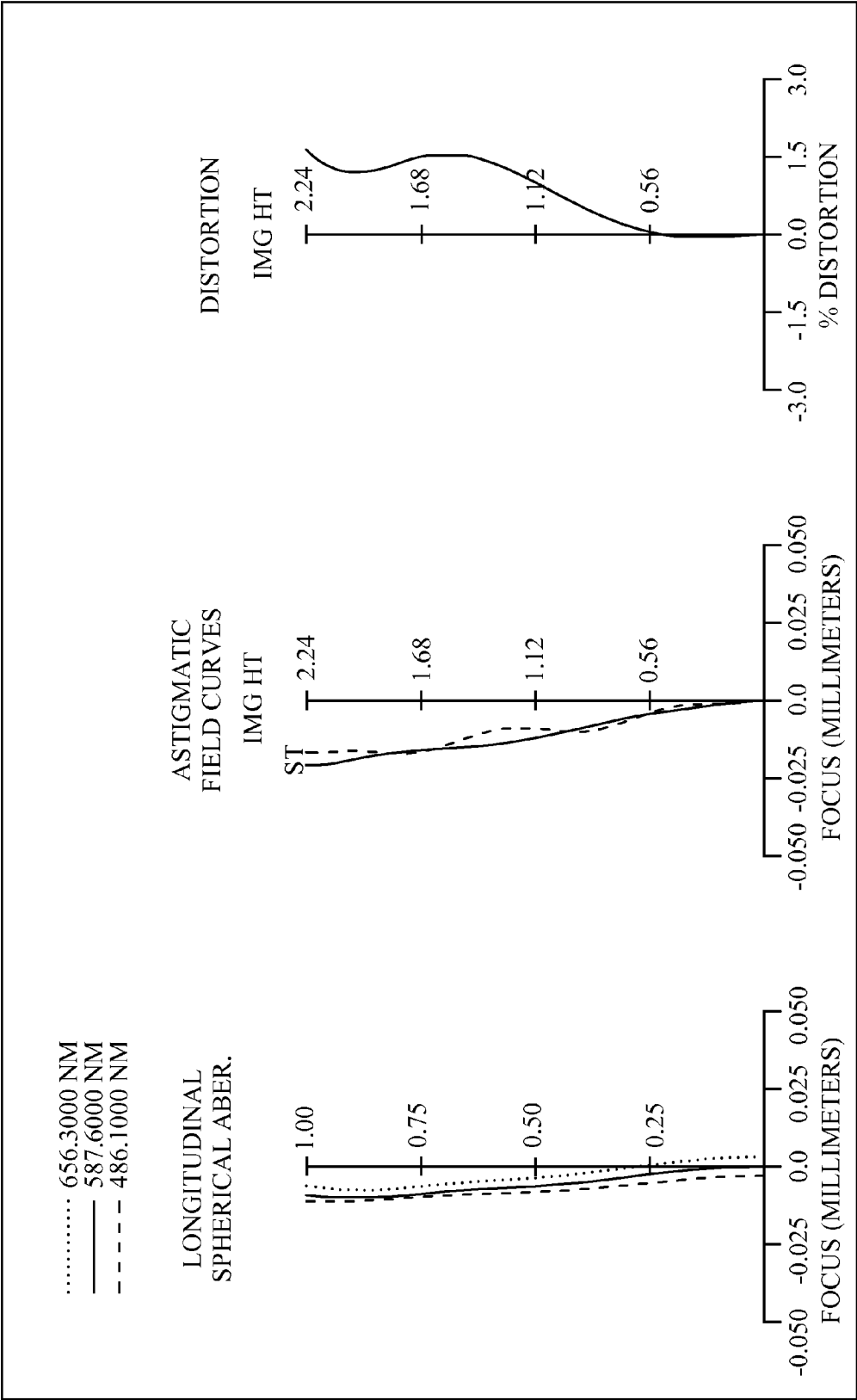


FIG. 1B

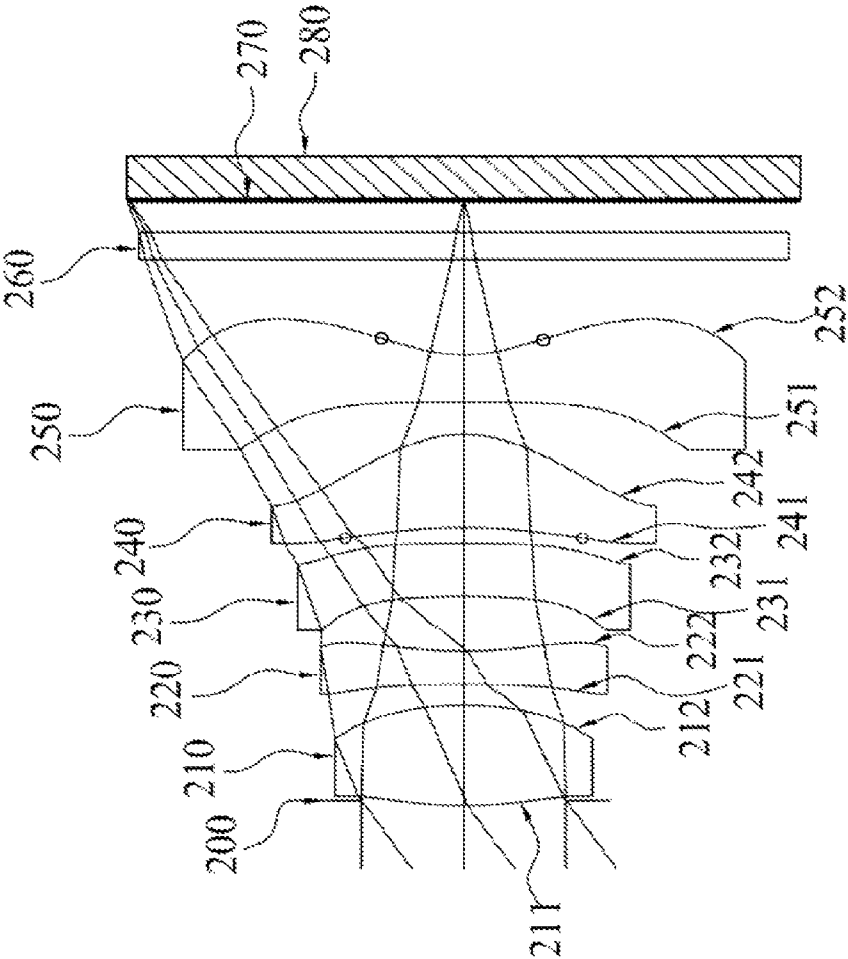


FIG. 2A

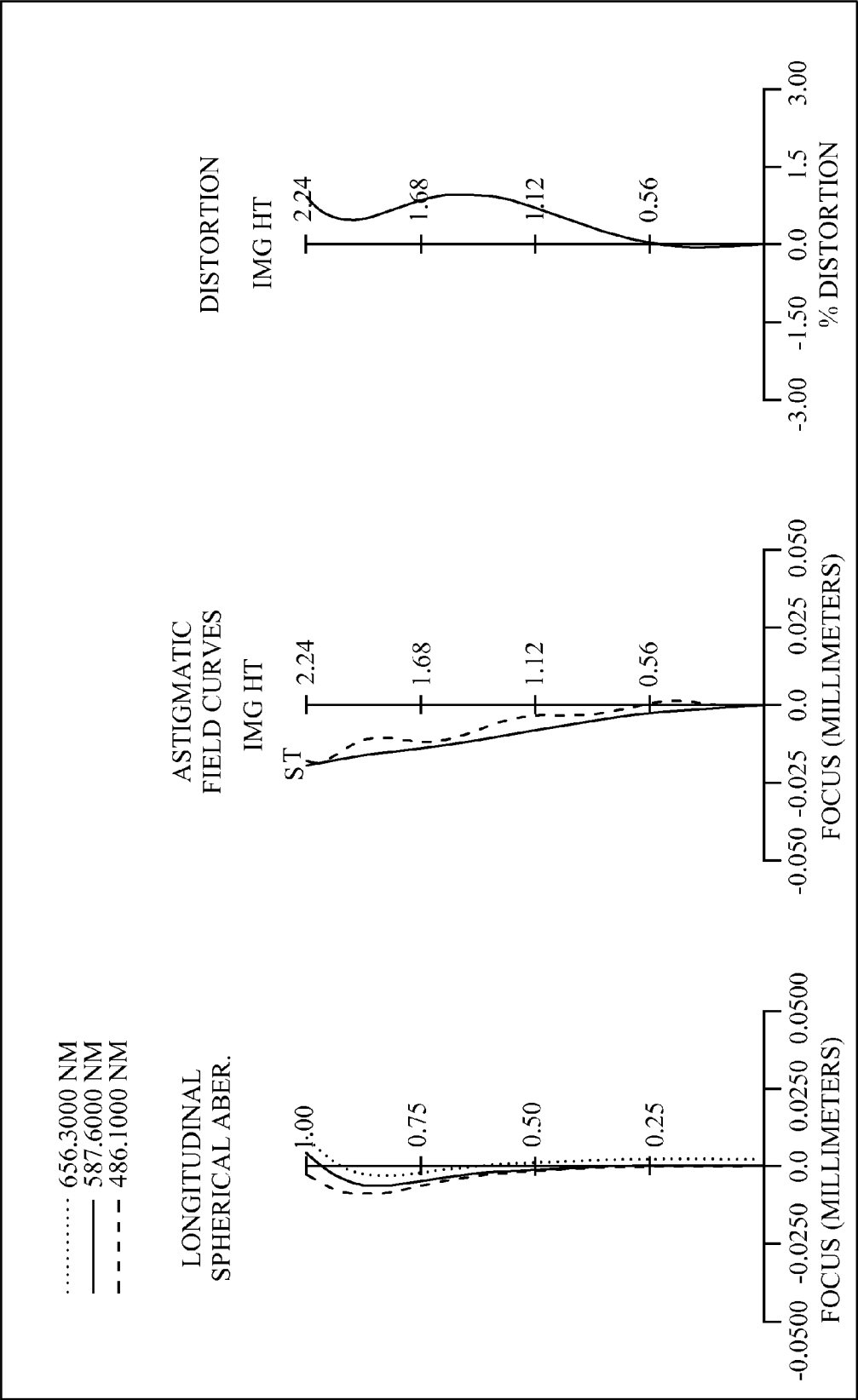


FIG. 2B

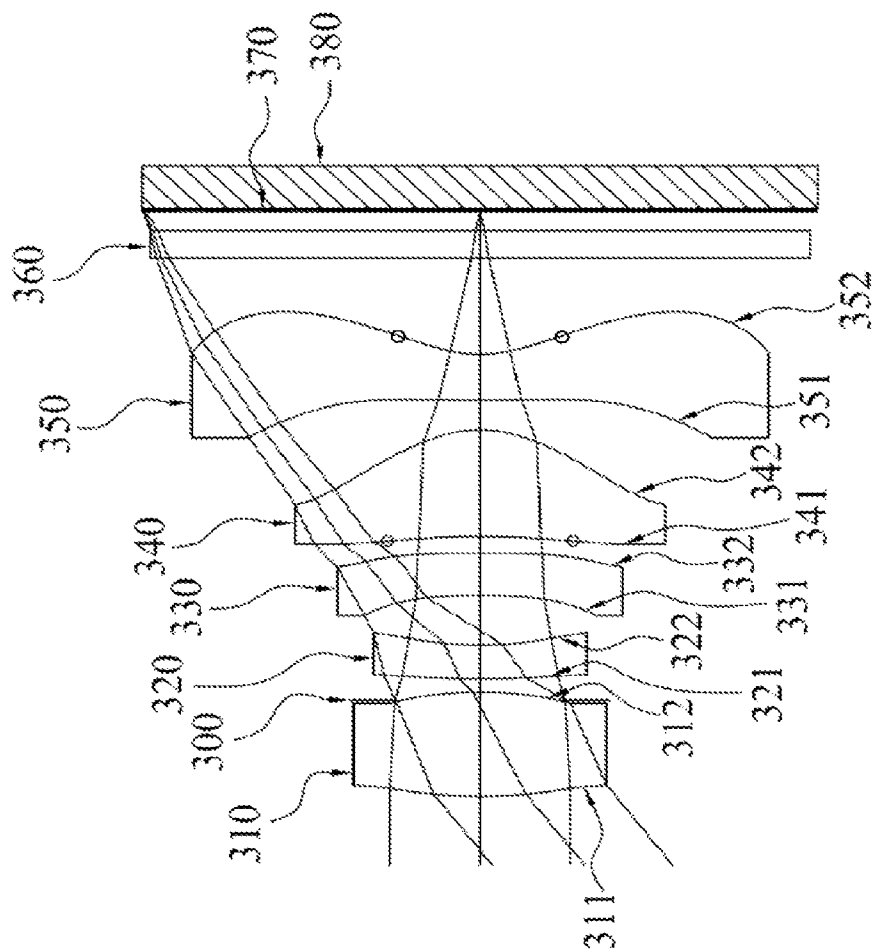
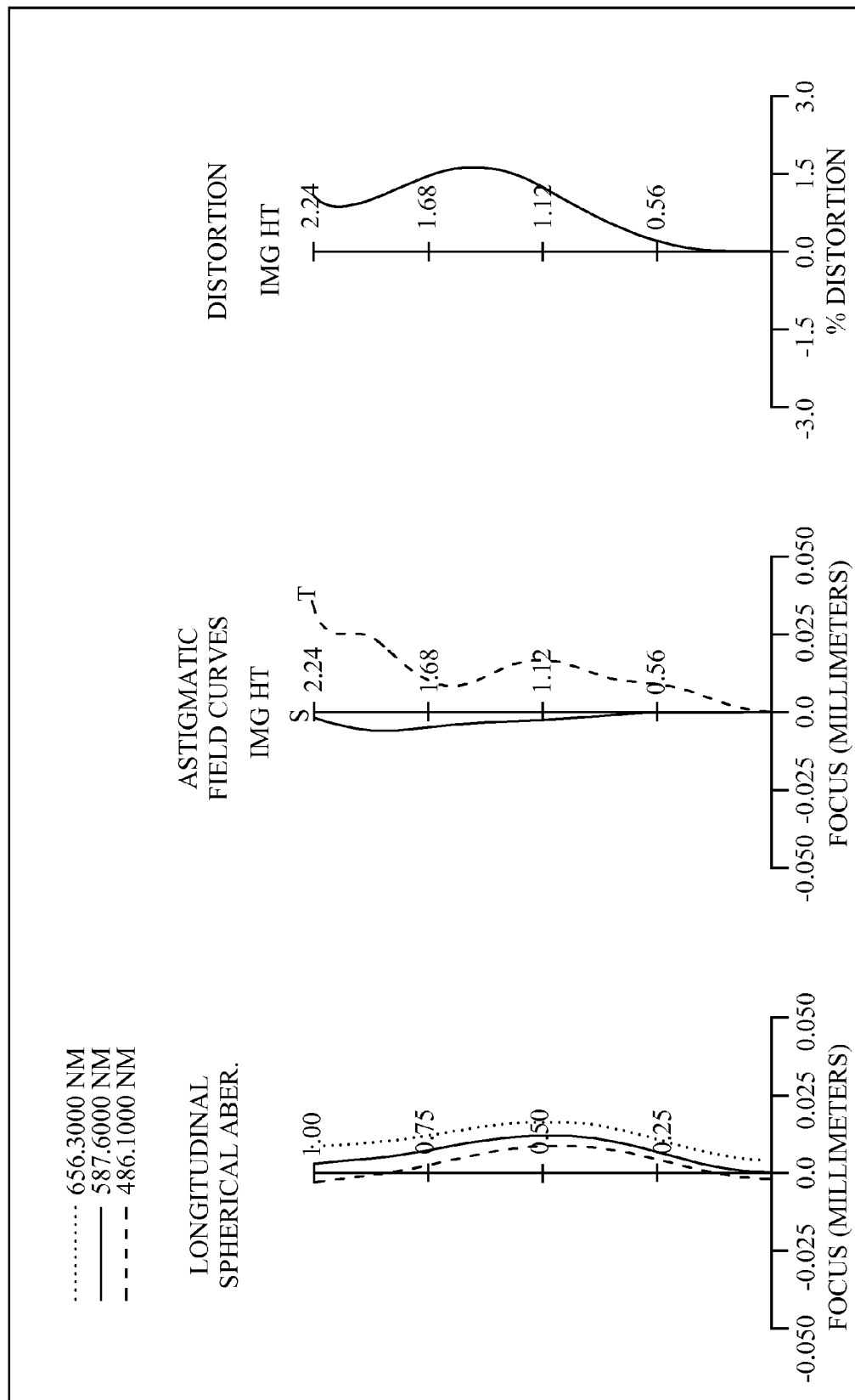


FIG. 3A



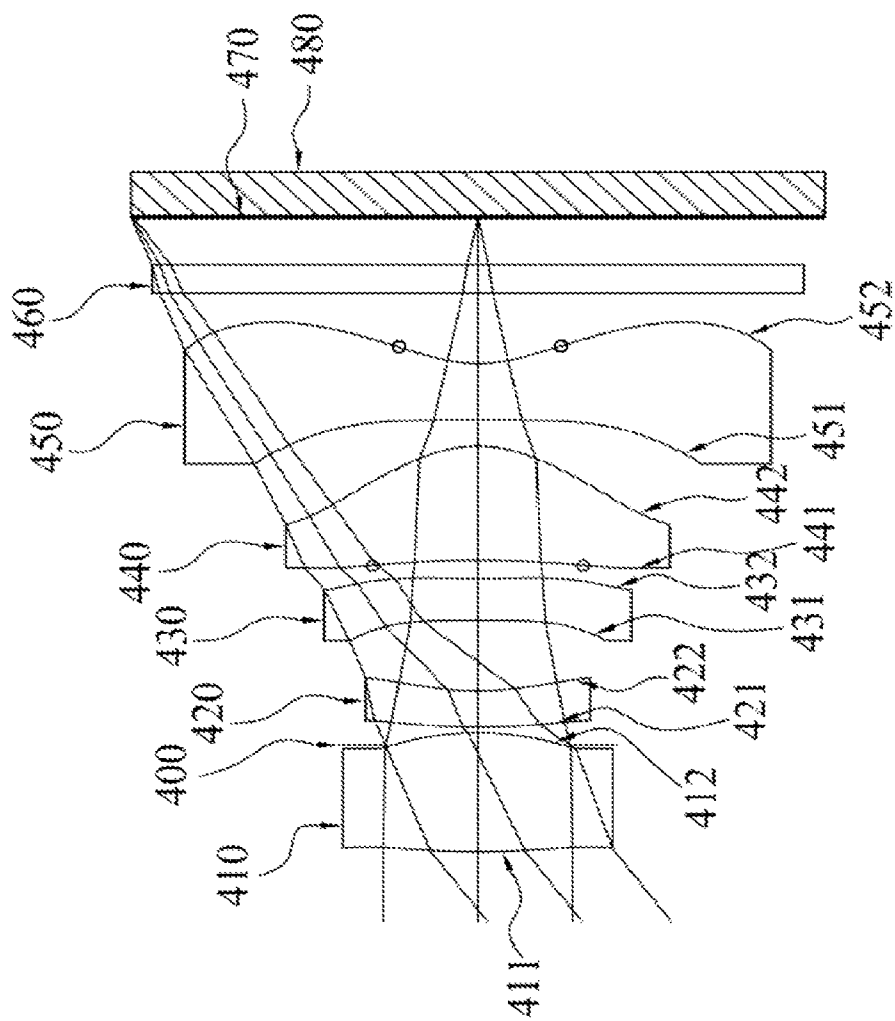


FIG. 4A

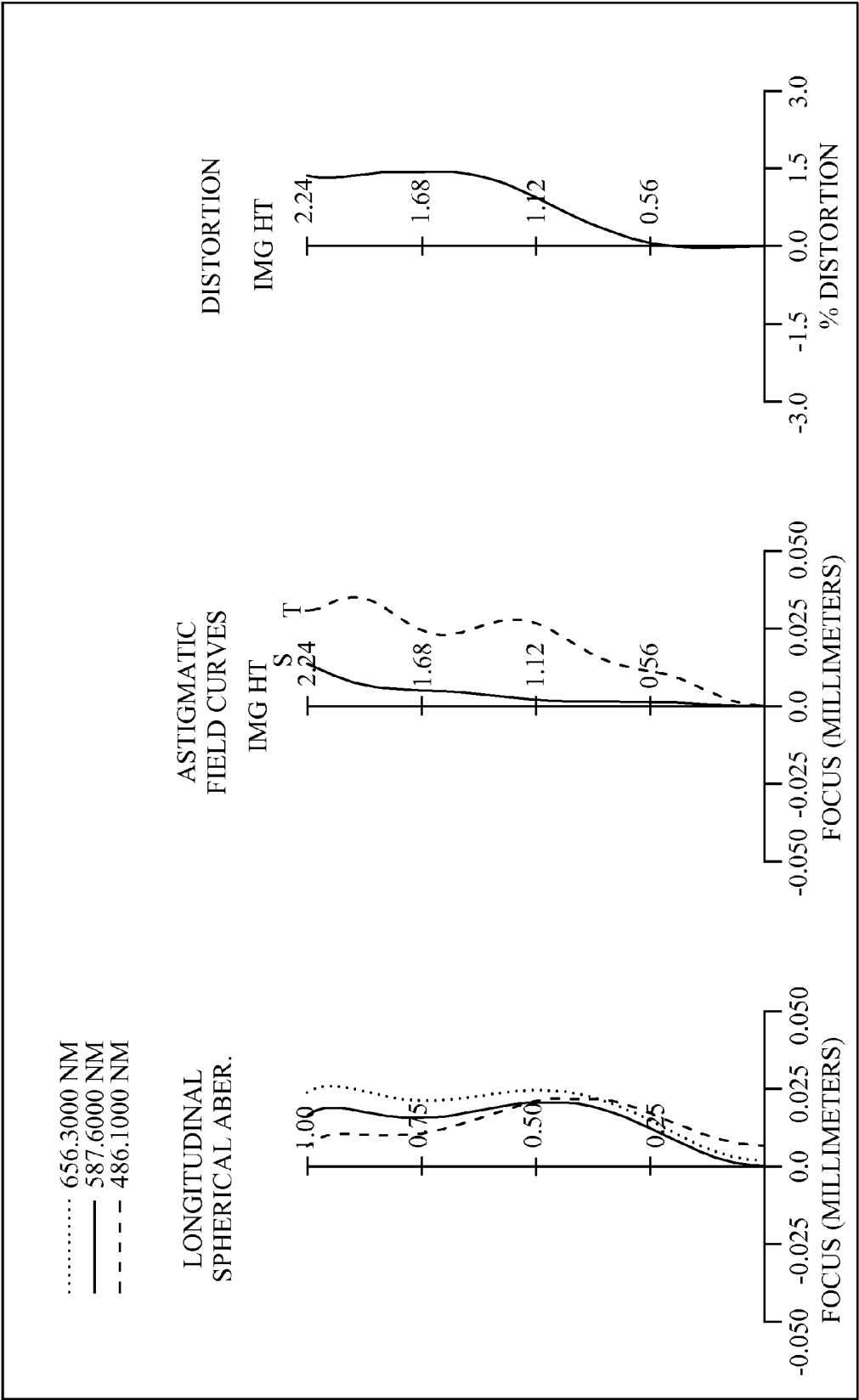


FIG. 4B

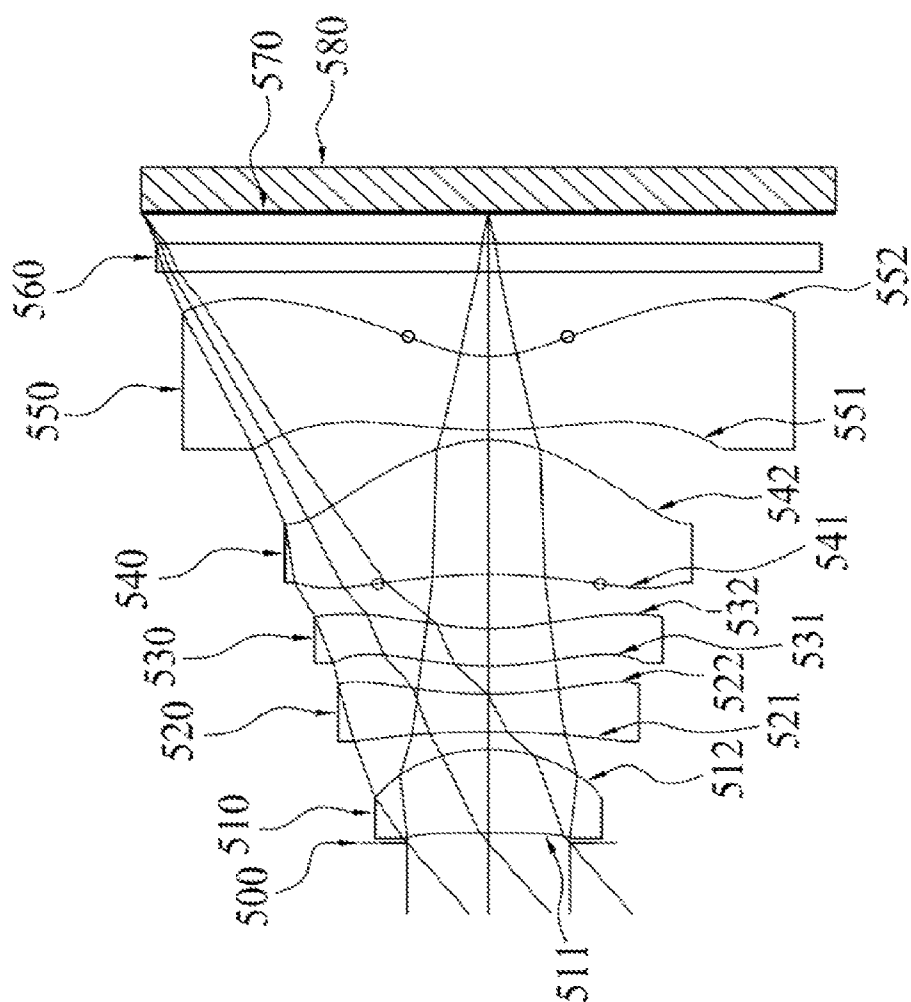


FIG. 5A

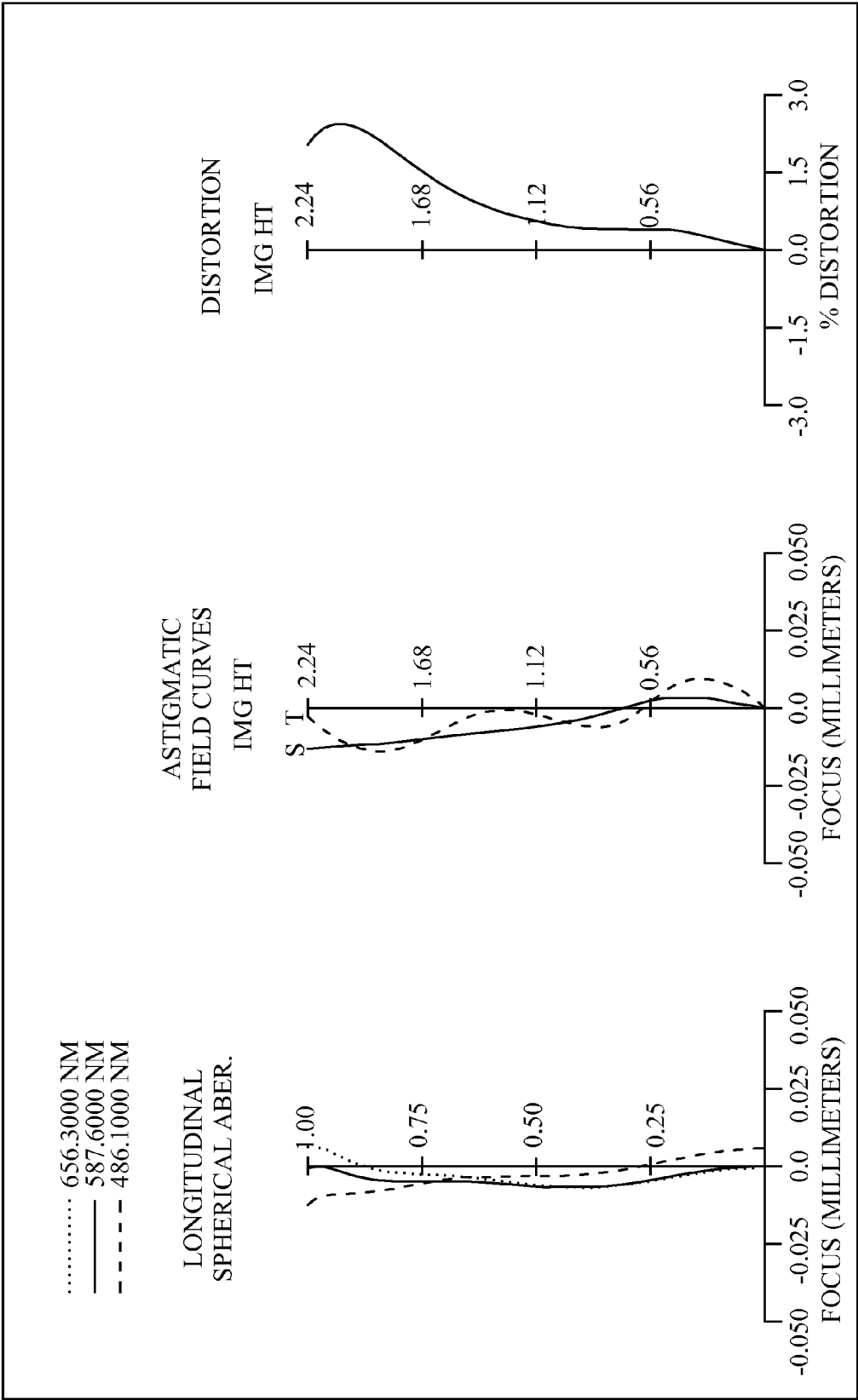


FIG. 5B

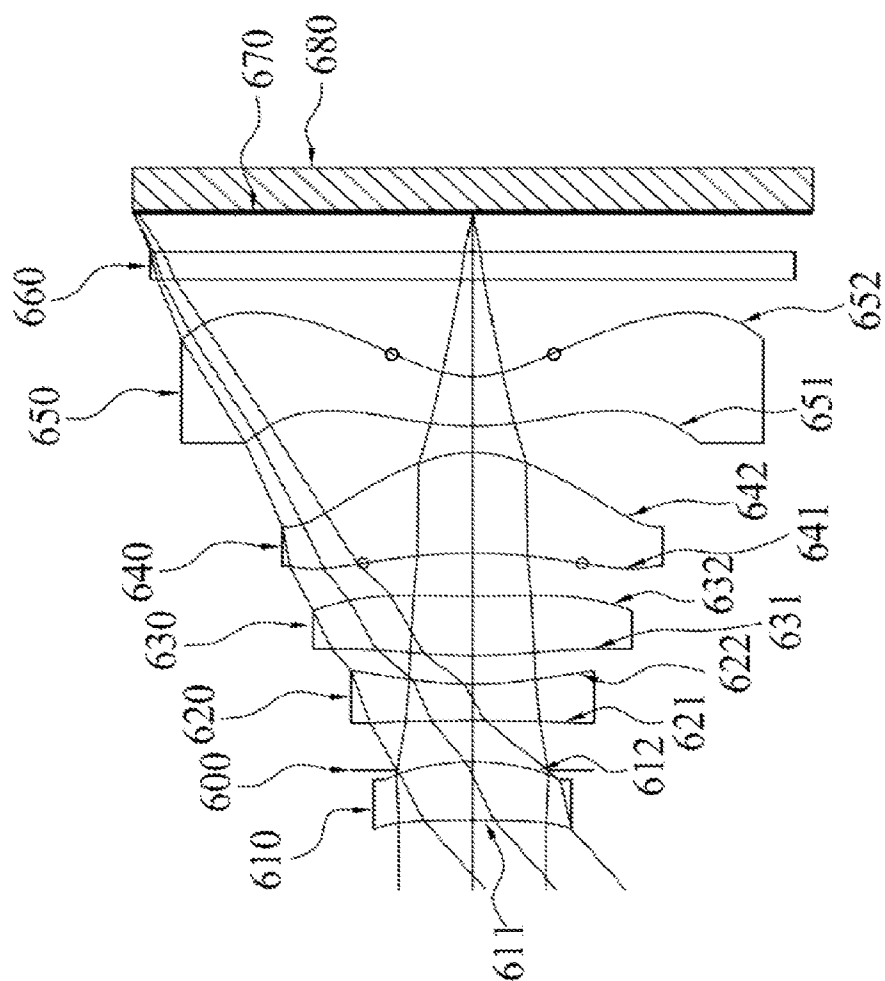


FIG. 6A

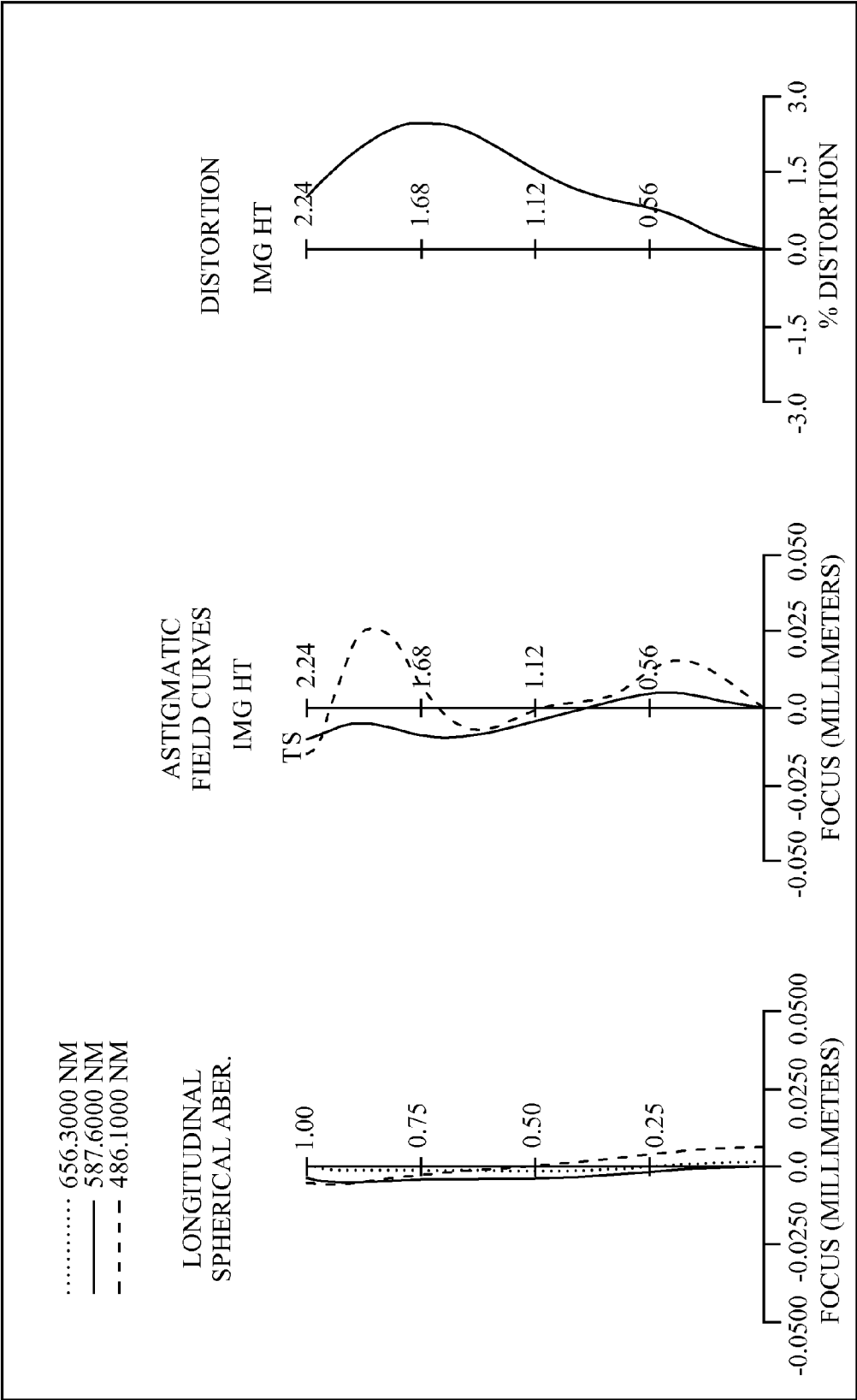


FIG. 6B

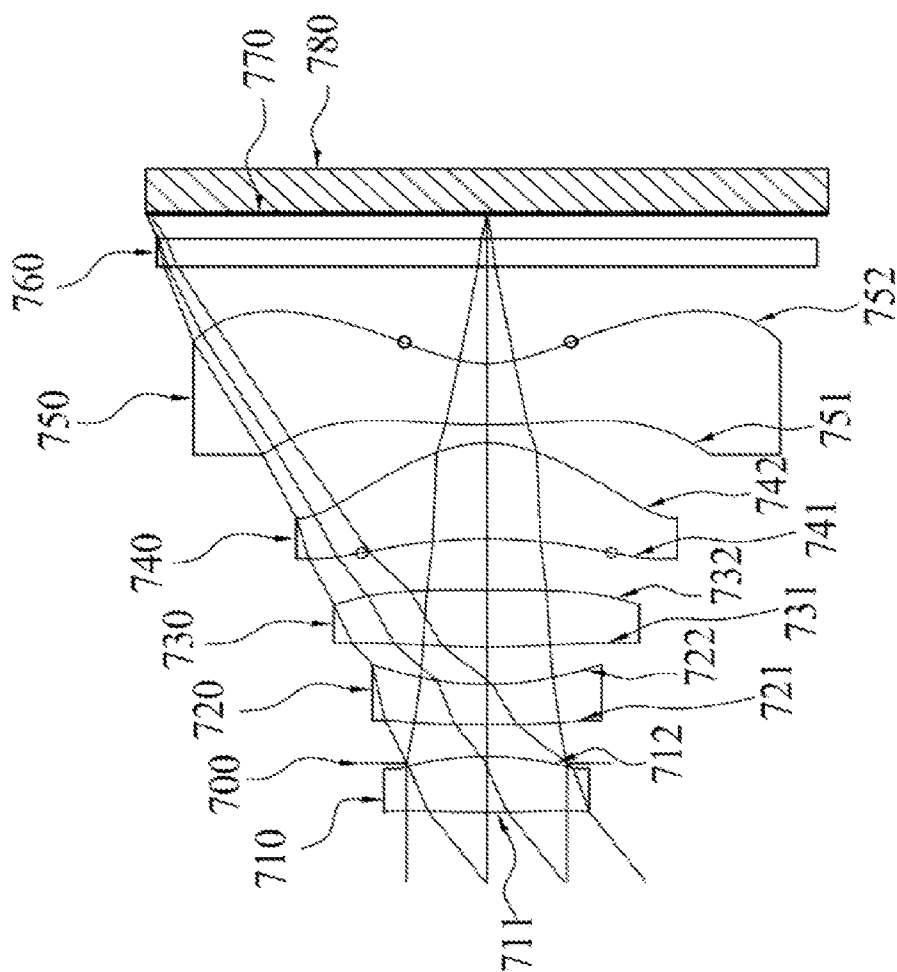


FIG. 7A

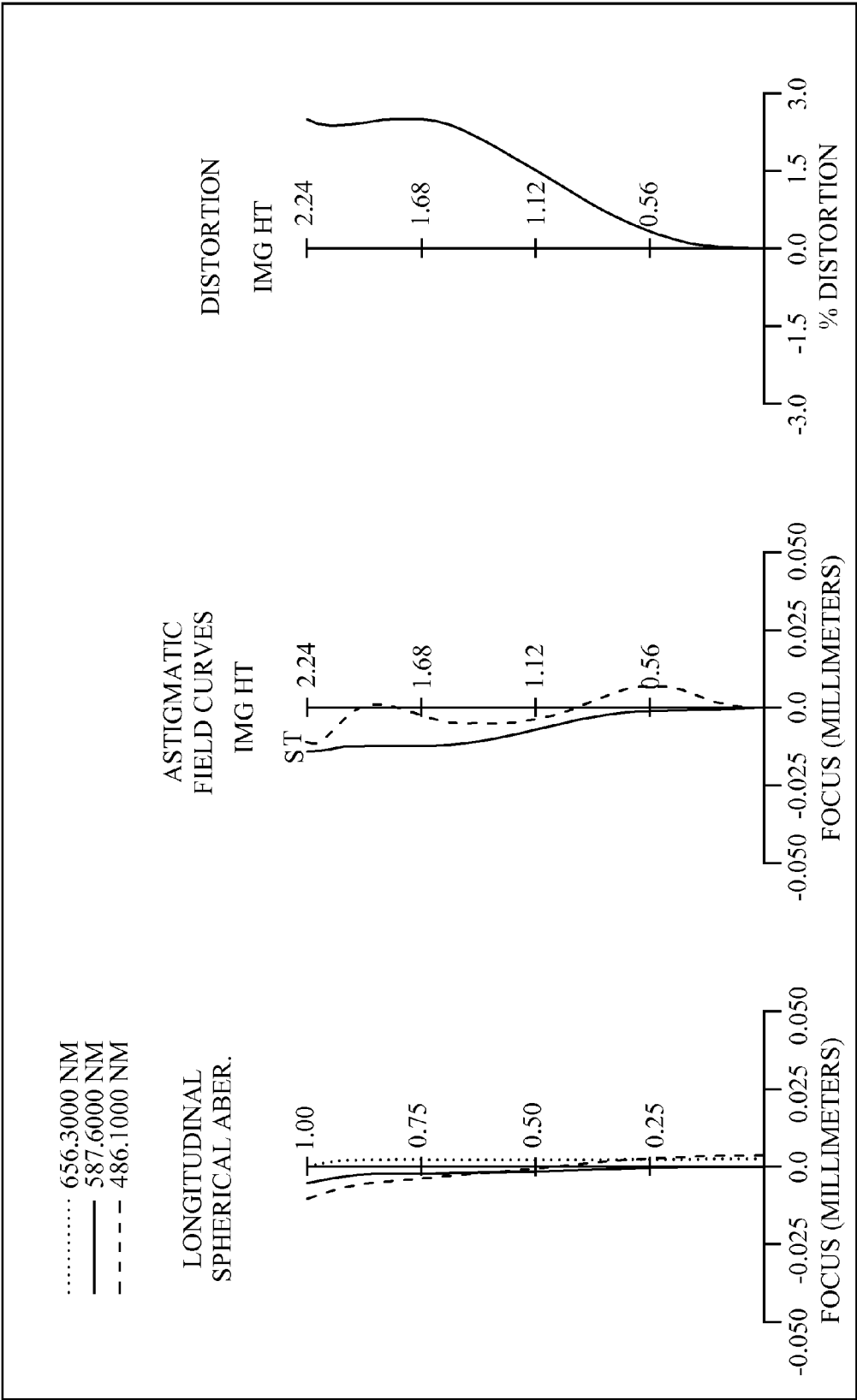


FIG. 7B

OPTICAL SYSTEM FOR IMAGING PICKUP

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to an optical system for imaging pickup, and more particularly to the optical system for imaging pickup to shorten total length and improve image quality for applying to electronic products.

2. Description of the Related Art

In compact electronic products such as digital still cameras or mobile phone cameras, an optical system for imaging pickup is generally installed for capturing images of an object, and the optical system for imaging pickup tends to be developed with a compact design and a low cost, while meeting the user requirements of good aberration correction ability, high resolution, and high image quality.

In general, a conventional optical system for imaging pickup of a compact electronic product comes with different designs, including the two-lens, three-lens, four-lens, and five-or-more lens designs. However, if the imaging quality is taken into consideration, the optical system for imaging pickup with the four-lens or five-lens designs has advantages on image aberration and modulation transfer function (MTF) performance, wherein the five-lens design having a higher resolution than the four-lens and five-lens designs thus being applicable for electronic products requiring the high quality and high pixels.

In various compact designs of the five-lens optical system for imaging pickup having a fixed focal length, the prior art adopts different combinations of positive and negative refractive powers as disclosed in U.S. Pat. No. 7,480,105, adopting the first lens element with negative refractive power and the second lens element with positive refractive power are used. Due to the two cemented glass lens elements, the manufacturing cost is higher which is unfavorable to the cost control in mass production. In addition, the glass lens element generally is thicker than the plastic lens element, and it is difficult to make an aspheric surface of the glass lens element and reduce the total length of the optical system. If the first lens element with negative refractive power is used, it will be more difficult to reduce the total length of the optical system.

In products such as compact digital cameras, web cameras, and mobile phone cameras, the optical system for imaging pickup requires a compact design, a short focal length, and a good aberration correction. As disclosed in U.S. Pat. No. 7,502,181, the fourth lens element with negative refractive power and the fifth lens element with negative refractive power are adopted to increase the back focal length and the total length of the optical imaging system, so that it is difficult to shorten the total length of the optical imaging system. In various five-lens designs of the optical system for imaging pickup with a fixed focal length, the first lens element has a convex image-side surface to facilitate a good balance between an increase of the view angle of the optical imaging system and the effect of the aberration correction, or the fourth lens element and the fifth lens element with different refractive powers are adopted, and the fourth lens element or the fifth lens element has an inflection point, but such optical system for imaging pickup fails to meet the requirements for a good aberration and a reduced total length of the optical imaging system. The present invention provides a more practical design for shortening the optical imaging system, adopting a combination of refractive powers of five lens elements, convex and concave optical surfaces. Wherein the fourth and fifth lens elements have positive refractive power and negative refractive power respectively, and this complementary

combination with a telecentric effect is favorable for reducing the back focal length and the total length of the optical system for imaging pickup effectively as well as further improving the image quality and applying the optical imaging lens assembly to compact electronic products.

SUMMARY OF THE INVENTION

Therefore, it is a primary objective of the present invention to provide an optical system for imaging pickup, sequentially arranged from an object side to an image side, comprising: the first lens element, the second lens element, the third lens element, the fourth lens element, and the fifth lens element, wherein, the first lens element with positive refractive power has a convex image-side surface; the second lens element with refractive power has a concave image-side surface; the third lens element has refractive power; the fourth lens element with refractive power has both object-side and image-side surfaces being aspheric; the fifth lens element with negative refractive power has a concave image-side surface, both object-side surface and image-side surface being aspheric, and at least one optical surface having at least one inflection point; and the following relations are satisfied:

$$|R_1|/R_2 < -1.0; \quad (1)$$

$$0.5 < f_1/f < 1.4; \quad (2)$$

$$-1.5 < f_5/f < -0.3; \quad (3)$$

wherein, R_1 is a curvature radius of the object-side surface of the first lens element, R_2 is a curvature radius of the image-side surface of the first lens element, f is a focal length of the optical system for imaging pickup, f_1 is a focal length of the first lens element, and f_5 is a focal length of the fifth lens element.

Moreover, the present invention provides an optical system for imaging pickup, as described above, further comprising an image sensor, wherein the second lens element has negative refractive power; the fourth lens element has a convex image-side surface; and the optical system for imaging pickup satisfies one or more of the following relations in addition to the relations (1), (2) and (3):

$$0.70 < S_D/T_D < 1.15; \quad (4)$$

$$|f_2/f_3| < 1.5; \quad (5)$$

$$0.6 < \text{ImgH}/f < 1.0; \quad (6)$$

$$-2.0 < R_2/R_4 < -0.3; \quad (7)$$

wherein, S_D is an axial distance between the stop and the image-side surface of the fifth lens element, T_D is an axial distance between the object-side surface of the first lens element and the image-side surface of the fifth lens element, f is the focal length of the optical system for imaging pickup, f_2 is a focal length of the second lens element, f_3 is a focal length of the third lens element, R_2 is a curvature radius of the image-side surface of the first lens element, R_4 is a curvature radius of the image-side surface of the second lens element, and ImgH is half of a diagonal length of an effective photo-sensitive area of the image sensor.

Moreover, the present invention provides an optical system for imaging pickup, as described above, wherein the first lens element has a convex object-side surface; the second lens element has negative refractive power; the fourth lens element has a concave object-side surface and a convex image-side surface; the fourth lens element and the fifth lens element can be made of plastic; and the optical system for imaging

pickup satisfies one or more of the following relations in addition to the relations (1), (2), (3) and (4):

$$0.3 < f_4/f < 0.9; \quad (8)$$

$$|R_1|/R_2 < -2.0; \quad (9)$$

$$|f_2/f_3| < 0.5; \quad (10)$$

$$3.0 < |f/f_4| + |f/f_5| < 5.0; \quad (11)$$

wherein, f is a focal length of the optical system for imaging pickup, f_2 is a focal length of the second lens element, f_3 is a focal length of the third lens element, f_4 is a focal length of the fourth lens element, f_5 is a focal length of the fifth lens element, R_1 is a curvature radius of the object-side surface of the first lens element, and R_2 is a curvature radius of the image-side surface of the first lens element.

Moreover, the present invention provides an optical system for imaging pickup, as described above, further comprising an image sensor at an image plane for imaging an imaged object, wherein the second lens element has negative refractive power; and the optical system for imaging pickup satisfies one or more of the following relations in addition to the relations (1), (2), (3) and (4):

$$TTL/ImgH < 2.1; \quad (12)$$

wherein TTL is the axial distance between the object-side surface of the first lens element and the image plane, and $ImgH$ is half of the diagonal length of an effective photosensitive area of the image sensor.

It is another objective of the present invention to provide an optical system for imaging pickup, sequentially arranged from an object side to an image side along an optical axis, comprising: the first lens element, the second lens element, the third lens element, the fourth lens element and the fifth lens element; wherein the first lens element with positive refractive power has a convex image-side surface; the second lens element with refractive power has a concave image-side surface; the third lens element with refractive power; the fourth lens element with positive refractive power has a convex image-side surface, both object-side surface and image-side surface being aspheric, and at least one of the object-side surface and the image-side surface has at least one inflection point; the fifth lens element with negative refractive power has a concave image-side surface and both object-side surface and image-side surface being aspheric, and at least one of the object-side surface and the image-side surface has at least one inflection point, and the optical system for imaging pickup satisfies the following relations:

$$|R_1|/R_2 < -1.0; \quad (1)$$

$$0.5 < f_1/f < 1.4; \quad (2)$$

$$-1.5 < f_2/f < -0.3; \quad (3)$$

wherein R_1 is the curvature radius of the object-side surface of the first lens element, R_2 is the curvature radius of the image-side surface of the first lens element, f is the focal length of the optical system for imaging pickup, f_1 is the focal length of the first lens element, and f_2 is the focal length of the second lens element.

Moreover, the present invention provides an optical system for imaging pickup as described above, wherein the first lens element has a convex object-side surface; the second lens element has negative refractive power; the fourth lens element and the fifth lens element are made of plastic; and the optical system for imaging pickup satisfies one or more of the following relations in addition to the relations (1), (2) and (3):

$$-2.0 < R_2/R_4 < -0.3; \quad (7)$$

$$|R_1|/R_2 < -2.0; \quad (9)$$

$$3.0 < |f/f_4| + |f/f_5| < 5.0; \quad (11)$$

wherein, R_1 is the curvature radius of the object-side surface of the first lens element, R_2 is the curvature radius of the image-side surface of the first lens element, R_4 is the curvature radius of the image-side surface of the second lens element, f is the focal length of the optical system for imaging pickup, f_4 is the focal length of the fourth lens element, and f_5 is the focal length of the fifth lens element.

Another objective of the present invention is to provide an optical system for imaging pickup, sequentially arranged from an object side to an image side, comprising: the first lens element, the second lens element, the third lens element, the fourth lens element and the fifth lens element; wherein the first lens element with positive refractive power has a convex image-side surface; the second lens element has negative refractive power; the third lens element has refractive power; the fourth lens element with positive refractive power both object-side surface and image-side surface being aspheric, and at least one of the object-side surface and the image-side surface has at least one inflection point; the fifth lens element with negative refractive power has a concave image-side surface and both object-side surface and image-side surface being aspheric, and at least one of the object-side surface and the image-side surface has at least one inflection point, and the optical system for imaging pickup satisfies the following relations:

$$|R_1|/R_2 < -1.0; \quad (1)$$

$$0.5 < f_1/f < 1.4; \quad (2)$$

$$-1.5 < f_2/f < -0.3; \quad (3)$$

$$0.35 < f_4/f < 0.67; \quad (13)$$

wherein, R_1 is the curvature radius of the object-side surface of the first lens element, R_2 is the curvature radius of the image-side surface of the first lens element, f is the focal length of the optical system for imaging pickup, f_1 is the focal length of the first lens element, f_2 is the focal length of the second lens element, and f_4 is the focal length of the fourth lens element, and f_5 is the focal length of the fifth lens element.

Moreover, the present invention provides an optical system for imaging pickup as described above, wherein the fourth lens element has a concave object-side surface and a convex image-side surface, and the optical system for imaging pickup satisfies one or more of the following relations in addition to the relations (1), (2), (3) and (4):

$$0.70 < S_D/T_D < 1.15; \quad (4)$$

$$|f_2/f_3| < 0.5; \quad (10)$$

$$-2.0 < R_2/R_4 < -0.3; \quad (7)$$

$$0.6 < ImgH/f < 1.0; \quad (6)$$

wherein, S_D is the axial distance between the stop and the image-side surface of the fifth lens element, T_D is the axial distance between the object-side surface of the first lens element and the image-side surface of the fifth lens element, f is the focal length of the optical system for imaging pickup, f_2 is the focal length of the second lens element, f_3 is the focal length of the third lens element, R_2 is the curvature radius of the image-side surface of the first lens element, R_4 is the curvature radius of the image-side surface of the second lens

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element, and ImgH is half of the diagonal length of an effective photosensitive area of the image sensor.

With the installation of the aforementioned first lens element, second lens element, third lens element, fourth lens element and fifth lens element with an appropriate interval apart from one another, the present invention can provide a good aberration correction and an advantageous modulation transfer function (MTF) in a greater field of view.

In the present invention of an optical system for imaging pickup comprising the first lens element, second lens element, third lens element, fourth lens element and fifth lens element, the first lens element with positive refractive power provides most of the refractive power required by the system, and the second lens element with negative refractive power can correct aberrations produced by the lens element with positive refractive power effectively and correct the Petzval sum of the system to make the image surface on the edge flatter. If the second lens element has a concave image-side surface, the negative refractive power of the second lens element can be enhanced appropriately according to the surface shape to provide a good aberration correction effect to the system. The third lens element corrects the aberration after the first lens element and the second lens element are assembled to improve the resolution of the optical system for imaging pickup, so that the overall aberration and distortion of the optical system for imaging pickup can meet the high resolution requirement. With the complementary fourth lens element with positive refractive power and fifth lens element with negative refractive power, the telephoto structure can facilitate reducing the back focal length, so as to shorten the total length. In addition, the fourth lens element having a convex image-side surface facilitates correcting the astigmatism and high-level aberration of the optical system for imaging pickup. The fifth lens element having a concave image-side surface can maintain the second principal point of the optical system for imaging pickup far from the image plane to facilitate reducing the total length of optical system for imaging pickup and achieve a compact system.

In the optical system for imaging pickup of the present invention, the arrangement of the stop produces a longer distance between the exit pupil of the optical system for imaging pickup and the image plane, so that the light of an image can be projected directly and then received by the image sensor to avoid dark corners or achieve the telecentric effect on the image side. In general, the telecentric effect can improve the brightness of the image plane and enhance the speed of receiving images by the CCD or CMOS image sensor.

In the optical system for imaging pickup of the present invention, the combination of the first lens element with positive refractive power, the second lens element with negative refractive power and the third lens element with refractive power, and the mutual compensation of the fourth lens element with refractive power and the fifth lens element with negative refractive power can reduce the total length of the optical system for imaging pickup effectively.

If the fifth lens element has an inflection point, the inflection point can be used for guiding light with an angle out from the edges of the fifth lens element, such that the light at the off-axis view angle is guided toward and received by the image sensor. In addition, the fourth lens element and the fifth lens element are made of plastic to facilitate the manufacture and lower the cost.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a schematic view of an optical system for imaging pickup in accordance with the first preferred embodiment of the present invention;

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FIG. 1B is a schematic view of a series of aberration curves of the first preferred embodiment of the present invention;

FIG. 2A is a schematic view of an optical system for imaging pickup in accordance with the second preferred embodiment of the present invention;

FIG. 2B is a schematic view of a series of aberration curves of the second preferred embodiment of the present invention;

FIG. 3A is a schematic view of an optical system for imaging pickup in accordance with the third preferred embodiment of the present invention;

FIG. 3B is a schematic view of a series of aberration curves of the third preferred embodiment of the present invention;

FIG. 4A is a schematic view of an optical system for imaging pickup in accordance with the fourth preferred embodiment of the present invention;

FIG. 4B is a schematic view of a series of aberration curves of the fourth preferred embodiment of the present invention;

FIG. 5A is a schematic view of an optical system for imaging pickup in accordance with the fifth preferred embodiment of the present invention;

FIG. 5B is a schematic view of a series of aberration curves of the fifth preferred embodiment of the present invention;

FIG. 6A is a schematic view of an optical system for imaging pickup in accordance with the sixth preferred embodiment of the present invention;

FIG. 6B is a schematic view of a series of aberration curves of the sixth preferred embodiment of the present invention;

FIG. 7A is a schematic view of an optical system for imaging pickup in accordance with the seventh preferred embodiment of the present invention; and

FIG. 7B is a schematic view of a series of aberration curves of the seventh preferred embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

With reference to FIG. 1A, an optical system for imaging pickup of the present invention, sequentially arranged from an object side to an image side, comprising: the first lens element **110**, the second lens element **120**, the third lens element **130**, the fourth lens element **140** and the fifth lens element **150**, wherein the first lens element **110** with positive refractive power has a convex image-side surface **112**; the second lens element **120** with refractive power has a concave image-side surface **122**; the third lens element **130** has refractive power; the fourth lens element **140** with refractive power has both object-side surface **141** and image-side surface **142** being aspheric, and at least one of the object-side surface **141** and image-side surface **142** has at least one inflection point (see small circles in FIG. 1A); and the fifth lens element **150** with negative refractive power has a concave image-side surface **152** and both object-side surface **151** and image-side surface **152** being aspheric, and at least one of the object-side surface and the image-side surface has at least one inflection point (see small circles in FIG. 1A). The optical system for imaging pickup further comprises a stop and an IR-filter **160**. More specifically, the stop can be an aperture stop **100** wherein it is a front aperture stop between an photographed object and the first lens element **110**, and the IR-filter **160** is between the fifth lens element **150** and an image plane **170** and generally made of panel glass without affecting the focal length of the optical system for imaging pickup of the present invention. The optical system for imaging pickup further comprises an image sensor **180** on the image plane **170** for imaging an object to be imaged. The aspheric surfaces of the first lens element **110**, second lens element **120**, third lens

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element **130**, fourth lens element **140**, and fifth lens element **150** comply with the aspherical surface formula as given in Equation (14).

$$X(Y) = \frac{(Y^2/R)}{1 + \sqrt{1 - (1 + K)(Y/R)^2}} + \sum_i (A_i) \cdot (Y^i) \quad (14)$$

Wherein,

X is the relative height between a point on the aspherical surface with a distance Y away from the optical axis and a tangent plane at the tip of the optical axis of the aspherical surface;

Y is the distance between a point on the curvature of the aspherical surface and the optical axis;

R is the curvature radius near the optical axis;

K is the conic coefficient; and

A_i is the i^{th} level aspherical surface coefficient.

In the optical system for imaging pickup of the present invention, the first lens element **110**, second lens element **120** and third lens element **130** can have spherical or aspheric surfaces. If aspheric optical surfaces are adopted, then the curvature radius of the optical surface can be used for changing the refractive power to reduce or eliminate aberrations, so as to shorten the total length of the optical imaging lens assembly effectively. With the installation of the first lens element **110**, second lens element **120**, third lens element **130**, fourth lens element **140** and fifth lens element **150**, the optical system for imaging pickup satisfies the relation (1) (2) and (3).

In the optical system for imaging pickup of the present invention, the positive refractive power is mainly provided by the first lens element **110**, such that if the ratio of the curvature radius R_1 of the object-side surface **111** of the first lens element **110** to the curvature radius R_2 of the image-side surface **112** of the first lens element **110** is limited according to the relation (1), the adjustment of the surface shape of the object-side surface **111** and the image-side surface **112** of the first lens element **110** can assist correcting the aberration of the system and facilitating allocating the positive refractive power of the first lens element **110**. If the relations (2) and (3) are satisfied, the design with the focal length f_1 of the first lens element **110** and the focal length f_5 of the fifth lens element **150** can be balanced within the range of specific conditions to facilitate allocating the refractive power of the first lens element **110** and the fifth lens element **150** appropriately to provide a suitable refractive power required by the optical system for imaging pickup and correct aberrations produced by the system, so as to adjust the back focal length and the total length appropriately.

If the relation (4) is satisfied, the position of the aperture stop relative to the distance between the first lens element **110** and the fifth lens element **150** can be adjusted to reduce the length of the optical system for imaging pickup. Similarly, if the relation (6) or (12) is satisfied, the total length of the optical system for imaging pickup can be reduced effectively, so that the image sensor **180** can have a larger effective pixel range within the same total length or specific focal length, so as to facilitate adjusting the view angle of the optical system for imaging pickup within an appropriate range.

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If the relation (5), (8) or (11) is satisfied, the focal length f_2 of the second lens element **120**, the focal length f_3 of the third lens element **130**, the focal length f_4 of the fourth lens element **140** and the focal length f_5 of the fifth lens element **150** can be balanced in a limited condition to appropriately allocate the refractive power required by the optical system for imaging pickup and reduce the sensitivity and aberration of the optical system for imaging pickup.

In the optical system for imaging pickup of the present invention, the first lens element **110** has a convex image-side surface **112**, and the second lens element **120** has a concave image-side surface **122**. If ratio of the curvature radii of the image-side surface **112** of the first lens element **110** and the image-side surface **122** of the second lens element **120** is limited according to the relation (7), the refractive power of the second lens element **120** and the positive refractive power of the first lens element **110** can be adjusted to complement each other, and the refractive power can be adjusted according to the curvature of the surface shape to facilitate correcting the aberration of the system.

The optical system for imaging pickup of the present invention is described by means of preferred embodiments with relevant drawings as follows.

First Preferred Embodiment

With reference to FIGS. 1A and 1B for a schematic view and a series of aberration curves of an optical system for imaging pickup in accordance with the first preferred embodiment of the present invention respectively, the optical system for imaging pickup, sequentially arranged from an object side to an image side along an optical axis, comprises: a stop, which can be an aperture stop **100**; the plastic first lens element **110** with positive refractive power having a convex object-side surface **111** and a convex image-side surface **112** and both object-side surface **111** and image-side surface **112** being aspheric; the plastic second lens element **120** with negative refractive power having a concave object-side surface **121** and a concave image-side surface **122**, and both object-side surface **121** and image-side surface **122** being aspheric; the plastic third lens element **130** with positive refractive power having a convex object-side surface **131** and a concave image-side surface **132**, and both object-side surface **131** and image-side surface **132** being aspheric; the plastic fourth lens element **140** with positive refractive power having a concave object-side surface **141** and a convex image-side surface **142**, and both object-side surface **141** and image-side surface **142** being aspheric; the plastic fifth lens element **150** with negative refractive power having a convex object-side surface **151** and a concave image-side surface **152**, and both object-side surface **151** and image-side surface **152** being aspheric, and both of the object-side surface **151** and image-side surface **152** having at least one inflection point; an IR-filter **160**, made of panel glass, without affecting the focal length of the optical lens assembly for image taking; and an image sensor **180** at an image plane **170**.

TABLE 1

Optical data of the first preferred embodiment $f = 3.24 \text{ mm}$, $F_{\text{no}} = 2.60$, $\text{HFOV} = 34.2 \text{ deg.}$						
Surface #		Curvature Radius	Thickness	Material	Index	Abbe # Focal length
0	Object	Plano		Infinity		
1	Ape. Stop	Plano	0.016			

TABLE 1-continued

Optical data of the first preferred embodiment f = 3.24 mm, Fno = 2.60, HFOV = 34.2 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
2	Lens 1	2.514710 (ASP)	0.683	Plastic	1.544	55.9	2.28
3		-2.217170 (ASP)	0.190				
4		-3.645200 (ASP)	0.265				
5	Lens 2	3.923000 (ASP)	0.297	Plastic	1.634	23.8	-2.94
6		4.465700 (ASP)	0.345				
7	Lens 3	5.716900 (ASP)	0.175	Plastic	1.634	23.8	29.07
8		-3.083000 (ASP)	0.699				
9	Lens 4	-0.737740 (ASP)	0.264	Plastic	1.544	55.9	1.61
10		6.676400 (ASP)	0.281				
11	Lens 5	0.737020 (ASP)	0.700	Plastic	1.530	55.8	-1.59
12		Plano	0.200				
13	IR-filter	Plano	0.292	Glass	1.517	64.2	—
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm. ASP stands for aspherical surfaces.

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The optical data of this preferred embodiment are listed in Table 1, wherein the object-side surfaces and the image-side surfaces of the first lens element **110** to the fifth lens element **150** comply with the aspherical surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 2 as follows:

TABLE 2

Aspheric coefficients of the first preferred embodiment					
Surface #	2	3	4	5	6
k =	-2.11293E+01	-2.55895E+00	2.93109E+00	-1.71103E+01	-1.52917E+01
A4 =	9.64131E-02	-1.08522E-01	-3.10754E-02	-4.00458E-02	-1.86538E-01
A6 =	-3.17417E-01	-1.04851E-01	-1.67141E-01	-6.14357E-02	-9.86904E-02
A8 =	3.19932E-01	-4.33036E-02	4.06091E-01	1.38123E-01	2.24541E-01
A10 =	-5.10215E-01	1.50080E-01	-6.98120E-01	-3.62984E-01	-3.25525E-01
A12 =	1.51467E-01	-1.51650E-01	9.50323E-01	3.81199E-01	1.20168E-01
A14 =	-2.44535E-02	-4.38450E-03	-5.18587E-01	-1.74955E-01	
Surface #	7	8	9	10	11
k =	-1.69619E+01	4.33083E+00	-3.44481E+00	-1.00000E+00	-5.04301E+00
A4 =	-1.27294E-01	6.46183E-02	-1.24756E-01	-1.08846E-01	-1.06817E-01
A6 =	-8.14955E-02	6.42169E-02	1.51132E-01	-2.60669E-02	3.56917E-02
A8 =	1.18636E-01	-2.16923E-01	-1.13255E-01	2.34237E-02	-1.22588E-02
A10 =	-9.00897E-02	3.05034E-01	6.82200E-02	-6.78874E-03	2.16080E-03
A12 =	3.86290E-02	-1.71392E-01	-1.65507E-02	-1.34988E-03	-2.02645E-04
A14 =		3.64181E-02	1.72306E-03	8.81944E-04	8.11767E-06

With reference to Table 1 and FIG. 1B for an optical system for imaging pickup of this preferred embodiment, the optical system for imaging pickup has a focal length f=3.24 (mm), an f-number Fno=2.60, and a half of maximum view angle HFOV=34.2°. After the optical data of this preferred embodiment are calculated and derived, the optical system for imaging pickup satisfies related conditions as shown in Table 3 below, and the related symbols have been described above and thus will not be described again.

TABLE 3

Data of related of the first preferred embodiment	
Relation	Data
$ R_1 /R_2$	-1.13
R_2/R_4	-0.57
f_1/f	0.70
$ f_2/f_3 $	0.10

TABLE 3-continued

Data of related of the first preferred embodiment	
Relation	Data
f_4/f	0.50
f_5/f	-0.49
$ f/f_4 + f/f_5 $	4.05
S_D/T_D	1.01
ImgH/f	0.69
TTL/ImgH	1.93

According to the optical data as shown in Table 1 and the series of aberration curves as shown in FIG. 1B, the optical system for imaging pickup in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

Second Preferred Embodiment

With reference to FIGS. 2A and 2B for a schematic view and a series of aberration curves of an optical system for imaging pickup in accordance with the second preferred embodiment of the present invention respectively, the optical system for imaging pickup, sequentially arranged from an object side to an image side along an optical axis, comprises:

a stop, which can be an aperture stop **200**; the plastic first lens element **210** with positive refractive power having a convex object-side surface **211** and a convex image-side surface **212** and both object-side surface **211** and image-side surface **212** being aspheric; the plastic second lens element **220** with negative refractive power having a concave object-side surface **221** and a concave image-side surface **222**, and both object-side surface **221** and image-side surface **222** being aspheric; the plastic third lens element **230** with negative refractive power having a concave object-side surface **231** and a convex image-side surface **232**, and both object-side surface **231** and image-side surface **232** being aspheric; the plastic fourth lens element **240** with positive refractive power having a concave object-side surface **241** and a convex image-side surface **242**, and both object-side surface **241** and image-side surface **242** being aspheric, and at least one of the object-side surface **241** and image-side surface **242** has at least one inflection point (see small circles in FIG. 2A); the plastic fifth lens element **250** with negative refractive power having a concave object-side surface **251** and a concave image-side surface **252**, and both object-side surface **251** and image-side surface **252** being aspheric, and the image-side surface **252** having at least one inflection point (see small circles in FIG. 2A); an IR-filter **260**, made of panel glass, without affecting the focal length of the optical lens assembly for image taking; and an image sensor **280** at an image plane **270**.

TABLE 4

Optical data of the second preferred embodiment f = 3.29 mm, Fno = 2.40, HFOV = 34.1 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.036				
2		2.604970 (ASP)	0.735				
3	Lens 1	-1.983380 (ASP)	0.150	Plastic	1.544	55.9	2.19
4		-12.097300 (ASP)	0.245				
5	Lens 2	2.697840 (ASP)	0.397	Plastic	1.634	23.8	-3.46
6		-9.518300 (ASP)	0.389				
7	Lens 3	-28.935200 (ASP)	0.119	Plastic	1.634	23.8	-22.55
8		-3.556100 (ASP)	0.685				
9	Lens 4	-0.743070 (ASP)	0.231	Plastic	1.544	55.9	1.59
10		-61.614300 (ASP)	0.347				
11	Lens 5	0.820630 (ASP)	0.700	Plastic	1.530	55.8	-1.53
12		Plano	0.200				
13	IR-filter	Plano	0.237	Glass	1.517	64.2	—
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm. ASP stands for aspherical surfaces.

The optical data of this preferred embodiment are listed in Table 4, wherein the object-side surfaces and the image-side surfaces of the first lens element **210** to the fifth lens element **250** comply with the aspherical surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 5 as follows:

TABLE 5

Aspheric coefficients of the second preferred embodiment					
Surface #	2	3	4	5	6
k =	-2.51219E+01	-6.18551E+00	3.00000E+00	-1.13175E+01	3.00000E+00
A4 =	1.03528E-01	-1.21947E-01	-6.42311E-02	-6.06786E-02	-1.84648E-01
A6 =	-3.21309E-01	-1.02647E-01	-1.30442E-01	-6.52542E-02	-1.31409E-01

TABLE 5-continued

Aspheric coefficients of the second preferred embodiment					
A8 =	2.89546E-01	5.89428E-02	4.23469E-01	1.53736E-01	2.75344E-01
A10 =	-3.28673E-01	1.71423E-02	-7.21205E-01	-3.94231E-01	-3.66226E-01
A12 =	4.67871E-02	-8.08678E-02	8.26780E-01	3.72224E-01	1.69011E-01
A14 =	-2.44535E-02	9.99005E-03	-3.88300E-01	-1.46766E-01	-1.33020E-11
Surface #	7	8	9	10	11
k =	-1.00000E+00	6.08615E+00	-3.36558E+00	-1.00000E+00	-5.91779E+00
A4 =	-1.11992E-01	3.79604E-02	-1.23868E-01	-3.87886E-02	-9.36696E-02
A6 =	-9.37303E-02	7.41536E-02	1.71181E-01	-6.03147E-02	3.14019E-02
A8 =	1.25484E-01	-2.14415E-01	-1.21236E-01	3.95880E-02	-1.09469E-02
A10 =	-6.61474E-02	2.99317E-01	6.68802E-02	-6.72586E-03	2.18529E-03
A12 =	4.57060E-02	-1.73542E-01	-1.67334E-02	-2.18669E-03	-2.87776E-04
A14 =	-9.90717E-03	3.74546E-02	1.06166E-03	7.09323E-04	1.54590E-05

With reference to Table 4 and FIG. 2B for an optical system for imaging pickup of this preferred embodiment, the optical system for imaging pickup has a focal length $f=3.29$ (mm), an f-number $Fno=2.40$, and a half of maximum view angle $HFOV=34.1^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical system for imaging pickup satisfies related conditions as shown in Table 6 below, and the related symbols have been described above and thus will not be described again.

TABLE 6

Data of related of the second preferred embodiment		
Relation	Data	
$ R_1 /R_2$	-1.31	
R_2/R_4	-0.74	
f_1/f	0.67	
$ f_2/f_3 $	0.15	
f_4/f	0.48	
f_5/f	-0.46	
$ f/f_4 + f/f_5 $	4.22	
S_D/T_D	0.99	
$ImgH/f$	0.68	
$TTL/ImgH$	1.95	

According to the optical data as shown in Table 4 and the series of aberration curves as shown in FIG. 2B, the optical system for imaging pickup in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

Third Preferred Embodiment

With reference to FIGS. 3A and 3B for a schematic view and a series of aberration curves of an optical system for imaging pickup in accordance with the third preferred embodiment of the present invention respectively, the optical system for imaging pickup, sequentially arranged from an object side to an image side along an optical axis, comprises: the plastic first lens element **310** with positive refractive power having a convex object-side surface **311** and a convex image-side surface **312** and both object-side surface **311** and image-side surface **312** being aspheric; a stop, which can be an aperture stop **300**; the plastic second lens element **320** with negative refractive power having a convex object-side surface **321** and a concave image-side surface **322**, and both object-side surface **321** and image-side surface **322** being aspheric; the plastic third lens element **330** with negative refractive power having a concave object-side surface **331** and a convex image-side surface **332**, and both object-side surface **331** and image-side surface **332** being aspheric; the plastic fourth lens element **340** with positive refractive power having a concave object-side surface **341** and a convex image-side surface **342**, and both object-side surface **341** and image-side surface **342** being aspheric, and at least one of the object-side surface **341** and image-side surface **342** has at least one inflection point (see small circles in FIG. 3A); the plastic fifth lens element **350** with negative refractive power having a convex object-side surface **351** and a concave image-side surface **352**, and both object-side surface **351** and image-side surface **352** being aspheric, and both of the object-side surface **351** and image-side surface **352** having at least one inflection point (see small circles in FIG. 3A); an IR-filter **360**, made of panel glass, without affecting the focal length of the optical lens assembly for image taking; and an image sensor **380** at an image plane **370**.

TABLE 7

Optical data of the third preferred embodiment $f = 2.93$ mm, $Fno = 2.45$, $HFOV = 37.0$ deg.						
Surface #		Curvature Radius	Thickness	Material	Index	Abbe # Focal length
0	Object	Plano	Infinity			
1	Lens 1	2.723040 (ASP)	0.759	Plastic	1.544	55.9 2.42
2		-2.307660 (ASP)	-0.057			
3	Ape. Stop	Plano	0.153	Plastic	1.650	21.4 -4.84
4		5.432500 (ASP)	0.245			
5	Lens 2	1.956380 (ASP)	0.361	Plastic	1.634	23.8 -10.81
6		-5.317500 (ASP)	0.307			
7	Lens 3	-24.271800 (ASP)	0.122			

TABLE 7-continued

Optical data of the third preferred embodiment f = 2.93 mm, Fno = 2.45, HFOV = 37.0 deg.						
Surface #		Curvature Radius	Thickness	Material	Index	Abbe # Focal length
8	Lens 4	-4.626000 (ASP)	0.773	Plastic	1.544	55.9 1.43
9		-0.704200 (ASP)	0.214			
10		11.216100 (ASP)	0.332			
11	Lens 5	0.725100 (ASP)	0.700	Plastic	1.530	55.8 -1.48
12		Plano	0.200			
13	IR-filter	Plano	0.154	Glass	1.517	64.2 —
14	Image	Plano	—			

Note:

Reference wavelength is 587.6 nm. ASP stands for aspherical surfaces.

The optical data of this preferred embodiment are listed in Table 7, wherein the object-side surfaces and the image-side surfaces of the first lens element **310** to the fifth lens element **350** comply with the aspherical surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 8 as follows:

TABLE 8

Aspheric coefficients of the third preferred embodiment					
Surface #	1	2	4	5	6
k =	-3.16677E+01	-1.44139E+01	-5.22091E+00	-7.75072E+00	-1.00000E+00
A4 =	1.24548E-01	-8.33728E-02	-6.61497E-03	-2.73383E-02	-2.69390E-01
A6 =	-2.83818E-01	-1.39288E-01	-1.32288E-01	-5.16571E-02	-1.37821E-01
A8 =	2.40834E-01	1.32753E-01	8.99122E-02	-7.20615E-03	3.97675E-01
A10 =	-1.88144E-01	6.48137E-03	-3.10314E-01	-2.67737E-01	-3.01206E-01
A12 =	4.86402E-02	-8.08674E-02	8.27201E-01	3.71524E-01	1.69579E-01
A14 =	-2.53314E-02	9.99028E-03	-3.88300E-01	-1.46766E-01	-7.70694E-05
Surface #	7	8	9	10	11
k =	-1.00000E+00	7.83609E+00	-3.37338E+00	-1.00000E+00	-5.14236E+00
A4 =	-1.48112E-01	5.89445E-02	-1.52098E-01	-7.41607E-02	-9.32363E-02
A6 =	-7.50650E-02	6.30700E-02	1.78676E-01	-5.00992E-02	3.20367E-02
A8 =	1.53121E-01	-2.20740E-01	-1.15286E-01	4.27537E-02	-1.14222E-02
A10 =	-4.98822E-02	2.99179E-01	6.81369E-02	-6.38496E-03	2.55215E-03
A12 =	5.17079E-02	-1.73208E-01	-1.71032E-02	-2.34084E-03	-3.45562E-04
A14 =	-8.50233E-03	3.72027E-02	3.76144E-04	5.86079E-04	1.29060E-05

With reference to Table 7 and FIG. 3B for an optical system for imaging pickup of this preferred embodiment, the optical system for imaging pickup has a focal length f=2.93 (mm), an f-number Fno=2.45, and a half of maximum view angle HFOV=37.0°. After the optical data of this preferred embodiment are calculated and derived, the optical system for imaging pickup satisfies related conditions as shown in Table 9 below, and the related symbols have been described above and thus will not be described again.

TABLE 9

Data of related relations of the third preferred embodiment	
Relation	Data
R ₁ /R ₂	-1.18
R ₂ /R ₄	-1.18
f ₁ /f	0.83
f ₂ /f ₃	0.45
f ₄ /f	0.49
f ₅ /f	-0.50
f ₂ f ₄ + f ₅	4.04

TABLE 9-continued

Data of related relations of the third preferred embodiment	
Relation	Data
S _D /T _D	0.78
ImgH/f	0.76
TTL/ImgH	1.87

According to the optical data as shown in Table 7 and the series of aberration curves as shown in FIG. 3B, the optical system for imaging pickup in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

Fourth Preferred Embodiment

With reference to FIGS. 4A and 4B for a schematic view and a series of aberration curves of an optical system for imaging pickup in accordance with the fourth preferred embodiment of the present invention respectively, the optical

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system for imaging pickup, sequentially arranged from an object side to an image side along an optical axis, comprises: the plastic first lens element **410** with positive refractive power having a convex object-side surface **411** and a convex image-side surface **412** and both object-side surface **411** and image-side surface **412** being aspheric; a stop, which can be an aperture stop **400**; the plastic second lens element **420** with negative refractive power having a convex object-side surface **421** and a concave image-side surface **422**, and both object-side surface **421** and image-side surface **422** being aspheric; the plastic third lens element **430** with negative refractive power having a concave object-side surface **431** and a concave image-side surface **432**, and both object-side surface **431** and image-side surface **432** being aspheric; the plastic fourth

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lens element **440** with positive refractive power having a concave object-side surface **441** and a convex image-side surface **442**, and both object-side surface **441** and image-side surface **442** being aspheric, and at least one of the object-side surface **441** and image-side surface **442** has at least one inflection point (see small circles in FIG. 4A); the plastic fifth lens element **450** with negative refractive power having a concave object-side surface **451** and a concave image-side surface **452**, and both object-side surface **451** and image-side surface **452** being aspheric, and the image-side surface **452** having at least one inflection point (see small circles in FIG. 4A); an IR-filter **460**, made of panel glass, without affecting the focal length of the optical lens assembly for image taking; and an image sensor **480** at an image plane **470**.

TABLE 10

Optical data of the fourth preferred embodiment f = 3.00 mm, Fno = 2.45, HFOV = 36.3 deg.						
Surface #	Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity			
1	4.740800 (ASP)	0.843				
2	Lens 1		Plastic	1.544	55.9	2.23
3	−1.529300 (ASP)	−0.088				
4	Ape. Stop	Plano				
5	4.227700 (ASP)	0.126				
6	Lens 2	0.256	Plastic	1.650	21.4	−4.26
7	1.631870 (ASP)	0.501				
8	−27.777800 (ASP)	0.294				
9	Lens 3		Plastic	1.650	21.4	−10.90
10	9.546100 (ASP)	0.132				
11	−4.793900 (ASP)	0.807				
12	Lens 4		Plastic	1.544	55.9	1.45
13	−0.716040 (ASP)	0.183				
14	−46.938000 (ASP)	0.398				
15	Lens 5		Plastic	1.530	55.8	−1.44
16	0.778550 (ASP)	0.500				
17	Plano	0.200				
18	IR-filter		Glass	1.517	64.2	—
19	Plano	0.338				
20	Image	Plano	—			

Note:

Reference wavelength is 587.6 nm. ASP stands for aspherical surfaces.

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The optical data of this preferred embodiment are listed in Table 10, wherein the object-side surfaces and the image-side surfaces of the first lens element **410** to the fifth lens element **450** comply with the aspherical surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 11 as follows:

TABLE 11

Aspheric coefficients of the fourth preferred embodiment					
Surface #	1	2	4	5	6
k =	−1.43485E+02	−1.24769E+01	−8.57062E+00	−1.45624E+01	−1.00000E+00
A4 =	7.02403E−02	−1.04406E−01	4.17329E−03	−2.89393E−02	−3.02041E−01
A6 =	−2.56188E−01	−9.41948E−02	−8.72953E−02	9.55910E−03	−1.46985E−01
A8 =	2.39831E−01	2.50360E−01	2.54476E−01	−8.37638E−02	4.30998E−01
A10 =	−1.51837E−01	−1.91942E−01	−6.74209E−01	−2.43424E−01	−3.56085E−01
A12 =	4.86385E−02	−8.08693E−02	8.27200E−01	3.71523E−01	1.69578E−01
A14 =	−2.53323E−02	9.98920E−03	−3.88301E−01	−1.46767E−01	−7.81456E−05
Surface #	7	8	9	10	11
k =	−1.00000E+00	1.26186E+01	−3.40910E+00	−1.00000E+00	−5.79898E+00
A4 =	−1.94948E−01	5.85946E−02	−1.53074E−01	−4.82332E−02	−8.26518E−02
A6 =	−6.25439E−02	6.25502E−02	1.80218E−01	−6.30240E−02	2.70388E−02
A8 =	1.49949E−01	−2.17264E−01	−1.15799E−01	4.25710E−02	−9.74785E−03
A10 =	−6.07251E−02	3.00952E−01	6.74532E−02	−5.26664E−03	2.57173E−03
A12 =	4.35824E−02	−1.73183E−01	−1.75116E−02	−2.04721E−03	−4.43064E−04
A14 =	−8.37219E−03	3.69773E−02	1.83301E−04	3.86326E−04	2.97028E−05

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With reference to Table 10 and FIG. 4B for an optical system for imaging pickup of this preferred embodiment, the optical system for imaging pickup has a focal length $f=3.00$ (mm), an f-number $Fno=2.45$, and a half of maximum view angle $HFOV=36.3^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical system for imaging pickup satisfies related conditions as shown in Table 12 below, and the related symbols have been described above and thus will not be described again.

TABLE 12

Data of related relations of the fourth preferred embodiment	
Relation	Data
$ R_1 /R_2$	-3.10
R_2/R_4	-0.94
f_1/f	0.74
$ f_2/f_3 $	0.39
f_4/f	0.48
f_5/f	-0.48
$ f_1/f_4 + f/f_5 $	4.16
S_D/T_D	0.78
$ImgH/f$	0.75
$TTL/ImgH$	1.97

According to the optical data as shown in Table 10 and the series of aberration curves as shown in FIG. 4B, the optical system for imaging pickup in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

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Fifth Preferred Embodiment

With reference to FIGS. 5A and 5B for a schematic view and a series of aberration curves of an optical system for imaging pickup in accordance with the fifth preferred embodiment of the present invention respectively, the optical system for imaging pickup, sequentially arranged from an object side to an image side along an optical axis, comprises: a stop, which can be an aperture stop **500**; the plastic first lens element **510** with positive refractive power having a concave object-side surface **511** and a convex image-side surface **512** and both object-side surface **511** and image-side surface **512** being aspheric; the plastic second lens element **520** with negative refractive power having a concave object-side surface **521** and a concave image-side surface **522**, and both object-side surface **521** and image-side surface **522** being aspheric; the plastic third lens element **530** with positive refractive power having a convex object-side surface **531** and a concave image-side surface **532**, and both object-side surface **531** and image-side surface **532** being aspheric; the plastic fourth lens element **540** with positive refractive power having a concave object-side surface **541** and a convex image-side surface **542**, and both object-side surface **541** and image-side surface **542** being aspheric, and at least one of the object-side surface **541** and image-side surface **542** has at least one inflection point (see small circles in FIG. 5A); the plastic fifth lens element **550** with negative refractive power having a convex object-side surface **551** and a concave image-side surface **552**, and both object-side surface **551** and image-side surface **552** being aspheric, and both of the object-side surface **551** and image-side surface **552** having at least one inflection point (see small circles in FIG. 5A); an IR-filter **560**, made of panel glass, without affecting the focal length of the optical lens assembly for image taking; and an image sensor **580** at an image plane **570**.

TABLE 13

Optical data of the fifth preferred embodiment $f = 2.74$ mm, $Fno = 2.60$, $HFOV = 38.8$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	0.072				
2		-50.000000 (ASP)	0.590				
3	Lens 1	-1.007210 (ASP)	0.125	Plastic	1.544	55.9	1.88
4		-5.967000 (ASP)	0.265				
5	Lens 2	2.204880 (ASP)	0.200	Plastic	1.634	23.8	-2.51
6		1.678780 (ASP)	0.268				
7	Lens 3	1.730380 (ASP)	0.383	Plastic	1.614	25.6	30.80
8	Lens 4	-3.189700 (ASP)	0.947	Plastic	1.544	55.9	1.64
9		-0.770610 (ASP)	0.070				
10	Lens 5	3.063200 (ASP)	0.522	Plastic	1.530	55.8	-1.87
11		0.703630 (ASP)	0.600				
12	IR-filter	Plano	0.200	Glass	1.517	64.2	—
13		Plano	0.219				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm. ASP stands for aspherical surfaces.

The optical data of this preferred embodiment are listed in Table 13, wherein the object-side surfaces and the image-side surfaces of the first lens element **510** to the fifth lens element **550** comply with the aspherical surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 14 as follows:

TABLE 14

Aspheric coefficients of the fifth preferred embodiment					
Surface #	2	3	4	5	6
k =	5.00000E+01	-1.53393E+00	2.18459E+00	-1.80052E+01	-8.43001E+00
A4 =	-2.64255E-01	-1.60376E-01	-1.18068E-02	-4.02985E-03	-1.37121E-01
A6 =	-4.07896E-01	-2.15120E-01	-1.17286E-01	-6.89948E-02	-7.54539E-02
A8 =	3.43635E-01	-1.13900E-01	4.53181E-01	1.19529E-01	2.38932E-01
A10 =	-1.89213E+00	1.71239E-01	-7.61441E-01	-3.51522E-01	-3.13697E-01
A12 =	-4.80222E-01	2.07620E-02	8.43063E-01	3.91536E-01	1.06673E-01
A14 =	-2.44535E-02	-9.47834E-01	-4.34136E-01	-1.73685E-01	
Surface #	7	8	9	10	11
k =	-7.77345E+00	4.53754E+00	-3.58026E+00	-1.00000E+00	-4.31611E+00
A4 =	-1.11116E-01	6.94257E-02	-1.52163E-01	-1.19058E-01	-8.58249E-02
A6 =	-5.38756E-02	6.61490E-02	1.45093E-01	-1.02667E-02	3.24380E-02
A8 =	1.22588E-01	-2.15497E-01	-1.13089E-01	2.39779E-02	-9.76448E-03
A10 =	-9.72693E-02	3.05921E-01	6.79795E-02	-6.66019E-03	1.91827E-03
A12 =	3.00711E-02	-1.71764E-01	-1.68199E-02	-1.48291E-03	-2.32832E-04
A14 =		3.55160E-02	2.06726E-03	6.41423E-04	1.04591E-05

With reference to Table 13 and FIG. 5B for an optical system for imaging pickup of this preferred embodiment, the optical system for imaging pickup has a focal length $f=2.74$ (mm), an f-number $Fno=2.60$, and a half of maximum view angle $HFOV=38.8^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical system for imaging pickup satisfies related conditions as shown in Table 15 below, and the related symbols have been described above and thus will not be described again.

TABLE 15

Data of related of the fifth preferred embodiment	
Relation	Data
$ R_1 /R_2$	-49.64
R_2/R_4	-0.46
f_1/f	0.69
$ f_2/f_3 $	0.08
f_4/f	0.60
f_5/f	-0.68
$ f/f_4 + f/f_5 $	3.14
S_D/T_D	1.02
$ImgH/f$	0.82
$TTL/ImgH$	1.93

According to the optical data as shown in Table 13 and the series of aberration curves as shown in FIG. 5B, the optical system for imaging pickup in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

Sixth Preferred Embodiment

With reference to FIGS. 6A and 6B for a schematic view and a series of aberration curves of an optical system for imaging pickup in accordance with the sixth preferred embodiment of the present invention respectively, the optical system for imaging pickup, sequentially arranged from an object side to an image side along an optical axis, comprises: the plastic first lens element **610** with positive refractive power having a concave object-side surface **611** and a convex image-side surface **612** and both object-side surface **611** and image-side surface **612** being aspheric; a stop, which can be an aperture stop **600**; the plastic second lens element **620** with negative refractive power having a concave object-side surface **621** and a concave image-side surface **622**, and both object-side surface **621** and image-side surface **622** being aspheric; the plastic third lens element **630** with positive refractive power having a convex object-side surface **631** and a concave image-side surface **632**, and both object-side surface **631** and image-side surface **632** being aspheric; the plastic fourth lens element **640** with positive refractive power having a concave object-side surface **641** and a convex image-side surface **642**, and both object-side surface **641** and image-side surface **642** being aspheric, and at least one of the object-side surface **641** and image-side surface **642** has at least one inflection point (see small circles in FIG. 6A); the plastic fifth lens element **650** with negative refractive power having a convex object-side surface **651** and a concave image-side surface **652**, and both object-side surface **651** and image-side surface **652** being aspheric, and both of the object-side surface **651** and image-side surface **652** having at least one inflection point (see small circles in FIG. 6A); an IR-filter **660**, made of panel glass, without affecting the focal length of the optical lens assembly for image taking; and an image sensor **680** at an image plane **670**.

TABLE 16

Optical data of the sixth preferred embodiment						
$f = 2.72$ mm, $Fno = 2.80$, $HFOV = 39.3$ deg.						
Surface #	Curvature	Radius	Thickness	Material	Index	Abbe # Focal length
0	Object	Plano	Infinity			
1		-18.181800 (ASP)	0.423			
2	Lens 1			Plastic	1.535	56.3 2.90
3		-1.438770 (ASP)	-0.060			
4	Ape. Stop	Plano	0.357			

TABLE 16-continued

Optical data of the sixth preferred embodiment f = 2.72 mm, Fno = 2.80, HFOV = 39.3 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
4	Lens 2	-12.500000 (ASP)	0.265	Plastic	1.650	21.4	-3.15
5		2.467840 (ASP)	0.205				
6		2.867950 (ASP)	0.430				
7	Lens 3	10.000000 (ASP)	0.311	Plastic	1.535	56.3	7.37
8		-3.081200 (ASP)	0.723				
9	Lens 4	-0.742190 (ASP)	0.191	Plastic	1.535	56.3	1.65
10		1.864760 (ASP)	0.358				
11	Lens 5	0.615860 (ASP)	0.700	Plastic	1.535	56.3	-1.91
12		Plano	0.200				
13	IR-filter	Plano	0.288	Glass	1.517	64.2	—
14	Image	Plano					

Note:

Reference wavelength is 587.6 nm. ASP stands for aspherical surfaces.

The optical data of this preferred embodiment are listed in Table 16, wherein the object-side surfaces and the image-side surfaces of the first lens element **610** to the fifth lens element **650** comply with the aspherical surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 17 as follows:

TABLE 17

Aspheric coefficients of the sixth preferred embodiment					
Surface #	1	2	4	5	6
k =	5.00000E+01	-2.62383E+00	3.00000E+00	-8.44258E+00	-2.00000E+01
A4 =	-1.83589E-01	-1.23340E-01	1.06835E-01	7.45311E-03	-1.10394E-01
A6 =	-1.27966E-01	-1.50107E-01	-3.45678E-01	-8.76993E-02	-5.61671E-02
A8 =	-1.10772E-01	-2.12846E-01	3.27273E-01	1.20798E-01	2.56874E-01
A10 =	-3.28810E-01	3.91032E-01	3.63783E-01	-2.12830E-01	-2.70683E-01
A12 =	1.03972E+00	2.80533E-01	-2.32682E+00	6.70465E-02	1.05955E-01
A14 =	-1.97349E+00	-1.04942E+00	2.63146E+00	7.92465E-02	
Surface #	7	8	9	11	10
k =	-3.43437E+00	4.35456E+00	-3.29210E+00	-3.22044E+00	-1.00000E+00
A4 =	-1.33788E-01	5.94254E-02	-1.29128E-01	-1.06229E-01	-1.66421E-01
A6 =	-7.57646E-02	6.96701E-02	1.34875E-01	4.01588E-02	-6.47463E-03
A8 =	1.36176E-01	-2.16913E-01	-1.08603E-01	-1.25954E-02	2.19366E-02
A10 =	-8.85539E-02	3.05901E-01	7.38133E-02	2.01455E-03	-7.03720E-03
A12 =	2.61340E-02	-1.70856E-01	-1.55639E-02	-1.60170E-04	-1.35853E-03
A14 =		3.59974E-02	6.35119E-04	1.40774E-06	7.08554E-04

With reference to Table 16 and FIG. 6B for an optical system for imaging pickup of this preferred embodiment, the optical system for imaging pickup has a focal length f=2.72 (mm), an f-number Fno=2.80, and a half of maximum view angle HFOV=39.3°. After the optical data of this preferred embodiment are calculated and derived, the optical system for imaging pickup satisfies related conditions as shown in Table 18 below, and the related symbols have been described above and thus will not be described again.

TABLE 18

Data of related relations of the sixth preferred embodiment	
Relation	Data
R ₁ /R ₂	-12.64
R ₂ /R ₄	-0.58

TABLE 18-continued

Data of related relations of the sixth preferred embodiment	
Relation	Data
f ₁ /f	1.07
f ₂ /f ₃	0.43
f ₄ /f	0.61
f ₅ /f	-0.70
f/f ₄ + f/f ₅	3.07
S _D /T _D	0.89
ImgH/f	0.83
TTL/ImgH	1.93

According to the optical data as shown in Table 16 and the series of aberration curves as shown in FIG. 6B, the optical system for imaging pickup in accordance with this preferred embodiment of the present invention provides good correc-

tion results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

Seventh Preferred Embodiment

With reference to FIGS. 7A and 7B for a schematic view and a series of aberration curves of an optical system for imaging pickup in accordance with the seventh preferred embodiment of the present invention respectively, the optical system for imaging pickup, sequentially arranged from an object side to an image side along an optical axis, comprises: the plastic first lens element **710** with positive refractive power having a convex object-side surface **711** and a convex image-side surface **712** and both object-side surface **711** and image-side surface **712** being aspheric; a stop, which can be an aperture stop **700**; the plastic second lens element **720** with negative refractive power having a convex object-side surface **721** and a concave image-side surface **722**, and both object-side surface **721** and image-side surface **722** being aspheric;

the plastic third lens element **730** with positive refractive power having a convex object-side surface **731** and a convex image-side surface **732**, and both object-side surface **731** and image-side surface **732** being aspheric; the plastic fourth lens element **740** with positive refractive power having a concave object-side surface **741** and a convex image-side surface **742**, and both object-side surface **741** and image-side surface **742** being aspheric, and at least one of the object-side surface **741** and image-side surface **742** has at least one inflection point (see small circles in FIG. 7A); the plastic fifth lens element **750** with negative refractive power having a convex object-side surface **751** and a concave image-side surface **752**, and both object-side surface **751** and image-side surface **752** being aspheric, and both of the object-side surface **751** and image-side surface **752** having at least one inflection point (see small circles in FIG. 7A); an IR-filter **760**, made of panel glass, without affecting the focal length of the optical lens assembly for image taking; and an image sensor **780** at an image plane **770**.

TABLE 19

Optical data of the seventh preferred embodiment f = 2.96 mm, Fno = 2.80, HFOV = 36.5 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	5.516400 (ASP)	0.400	Plastic	1.544	55.9	2.85
2		-2.101140 (ASP)	-0.045				
3	Ape. Stop	Plano	0.277	Plastic	1.650	21.4	-3.63
4		9.529900 (ASP)	0.290				
5	Lens 2	1.867120 (ASP)	0.267	Plastic	1.543	56.5	7.78
6		4.309500 (ASP)	0.411				
7	Lens 3	-222.005800 (ASP)	0.368	Plastic	1.514	56.8	1.68
8		-3.026700 (ASP)	0.690				
9	Lens 4	-0.723270 (ASP)	0.130	Plastic	1.530	55.8	-1.62
10		3.716900 (ASP)	0.440				
11	Lens 5	0.669090 (ASP)	0.700	Plastic	1.517	64.2	—
12		Plano	0.200				
13	IR-filter	Plano	0.181	Glass			
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm. ASP stands for aspherical surfaces.

The optical data of this preferred embodiment are listed in Table 19, wherein the object-side surfaces and the image-side surfaces of the first lens element **710** to the fifth lens element **750** comply with the aspherical surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 20 as follows:

TABLE 20

Aspheric coefficients of the seventh preferred embodiment					
Surface #	1	2	4	5	6
k =	4.79685E+01	-6.66867E+00	3.00000E+00	-4.30281E+00	-8.58192E+00
A4 =	-1.64173E-01	-7.91033E-02	1.02530E-01	3.20005E-02	-1.20354E-01
A6 =	-5.43124E-02	-1.52212E-01	-2.63808E-01	-5.90405E-02	-5.01619E-02
A8 =	-3.54434E-01	-5.85099E-02	3.43161E-01	1.31413E-01	2.50253E-01
A10 =	7.95098E-02	1.82446E-01	3.53869E-01	-2.11324E-01	-2.55920E-01
A12 =	1.03152E+00	2.81019E-01	-2.32687E+00	6.34594E-02	1.19385E-01
A14 =	-1.97355E+00	-1.04946E+00	2.63142E+00	1.17520E-01	

TABLE 20-continued

Aspheric coefficients of the seventh preferred embodiment					
Surface #	7	8	9	11	10
k =	-2.00000E+01	4.45444E+00	-3.42382E+00	-4.35640E+00	-1.00000E+00
A4 =	-9.17196E-02	4.46124E-02	-1.26749E-01	-9.41956E-02	-1.44380E-01
A6 =	-9.95546E-02	6.65981E-02	1.33660E-01	3.59961E-02	1.64936E-03
A8 =	1.37767E-01	-2.15742E-01	-1.07660E-01	-1.13323E-02	2.14288E-02
A10 =	-7.45911E-02	3.07424E-01	7.40354E-02	1.85296E-03	-7.18107E-03
A12 =	2.56752E-02	-1.70631E-01	-1.60621E-02	-1.79998E-04	-1.31822E-03
A14 =		3.58601E-02	5.52373E-05	7.01791E-06	7.22255E-04

With reference to Table 19 and FIG. 7B for an optical system for imaging pickup of this preferred embodiment, the optical system for imaging pickup has a focal length $f=2.96$ (mm), an f-number $Fno=2.80$, and a half of maximum view angle $HFOV=36.5^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical system for imaging pickup satisfies related conditions as shown in Table 21 below, and the related symbols have been described above and thus will not be described again.

TABLE 21

Data of related relations of the seventh preferred embodiment	
Relation	Data
$ R_1 /R_2$	-2.63
R_2/R_4	-1.13
f_1/f	0.96
$ f_2/f_3 $	0.47
f_4/f	0.57
f_5/f	-0.55
$ f/f_4 + f/f_5 $	3.58
S_D/T_D	0.89
$ImgH/f$	0.76
$TTL/ImgH$	1.89

According to the optical data as shown in Table 19 and the series of aberration curves as shown in FIG. 7B, the optical system for imaging pickup in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

In the optical system for imaging pickup of the present invention, the lens elements can be made of glass or plastic material. If the lens elements are made of glass, the freedom for distributing the refractive power of the optical system for imaging pickup can be increased. If plastic material is adopted to produce the lens elements, the production cost will be reduced effectively. Additionally, the surfaces of the lens elements can be aspheric and easily made into non-spherical profiles, allowing more design parameter freedom which can be used to reduce aberrations and the number of the lens elements used in an optical system. Consequently, the total track length of the image capturing lens assembly can be effectively reduced.

In the optical system for imaging pickup of the present invention, if the lens element has a convex surface, then the surface of the lens element is convex at a position in proximity to the optical axis; and if the lens element has a concave surface, then the surface of the lens element is concave at a position in proximity to the optical axis.

In the optical system for imaging pickup of the present invention, at least one stop such as a glare stop or a field stop is provided for reducing stray lights to improve the image quality, limiting the image field size, or other functionalities.

At least one stop can be positioned in front of the first lens element, between lens elements, or before the image plane of the optical system for imaging pickup according to the preference of the optical designer. Additionally, the optical system for imaging pickup can be utilized in 3D (three-dimensional) applications.

In the optical system for imaging pickup of the present invention, the aperture stop can be a front or middle aperture stop. If the aperture stop is a front aperture stop, a longer distance between the exit pupil of the optical system for imaging pickup and the image plane can be achieved to provide the telecentric effect and improve the image receiving efficiency of the image sensor such as the CCD or CMOS. If the aperture stop is a middle aperture stop, the view angle of the system can be expanded to provide a wide angle lens of the optical system for imaging pickup.

Tables 1 to 21 show changes of values of an optical imaging lens assembly in accordance with different preferred embodiments of the present invention respectively, and even if different values are used, products of the same structure are intended to be covered by the scope of the present invention. It is noteworthy to point out that the aforementioned description and the illustration of related drawings are provided for the purpose of explaining the technical characteristics of the present invention, but not intended for limiting the scope of the present invention.

What is claimed is:

1. An optical system for imaging pickup, sequentially arranged from an object side to an image side, comprising:
 - a first lens element with positive refractive power, having a convex image-side surface;
 - a second lens element with refractive power, having a concave image-side surface;
 - a third lens element with refractive power;
 - a fourth lens element with refractive power, having both object-side surface and image-side surface being aspheric; and
 - a fifth lens element with negative refractive power, having a concave image-side surface, both object-side surface and image-side surface being aspheric, and at least one of the object-side surface and the image-side surface having at least one inflection point; wherein, R_1 is a curvature radius of the object-side surface of the first lens element, R_2 is a curvature radius of the image-side surface of the first lens element, f is a focal length of the optical system for imaging pickup, f_1 is a focal length of the first lens element, f_5 is a focal length of the fifth lens element, and the following relations are satisfied:

$$|R_1|/R_2 < -1.0;$$

$$0.5 < f_1/f < 1.4; \text{ and}$$

$$-1.5 < f_5/f < -0.3.$$

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2. The optical system for imaging pickup of claim 1, further comprising a stop, wherein S_D is an axial distance between the stop and the image-side surface of the fifth lens element, T_D is an axial distance between the object-side surface of the first lens element and the image-side surface of the fifth lens element, and the following relation is satisfied:

$$0.70 < S_D/T_D < 1.15.$$

3. The optical system for imaging pickup of claim 2, wherein the second lens element has negative refractive power.

4. The optical system for imaging pickup of claim 3, wherein the fourth lens element has a convex image-side surface.

5. The optical system for imaging pickup of claim 4, wherein f_2 is a focal length of the second lens element, f_3 is a focal length of the third lens element, and the following relation is satisfied:

$$|f_2/f_3| < 1.5.$$

6. The optical system for imaging pickup of claim 5, further comprising an image sensor, wherein ImgH is half of a diagonal length of an effective photosensitive area of the image sensor, f is the focal length of the optical system for imaging pickup, and the following relation is satisfied:

$$0.6 < \text{ImgH}/f < 1.0.$$

7. The optical system for imaging pickup of claim 5, wherein R_2 is the curvature radius of the image-side surface of the first lens element, R_4 is a curvature radius of the image-side surface of the second lens element, and the following relation is satisfied:

$$-2.0 < R_2/R_4 < -0.3.$$

8. The optical system for imaging pickup of claim 3, wherein f_4 is a focal length of the fourth lens element, f is the focal length of the optical system for imaging pickup, and the following relation is satisfied:

$$0.3 < f_4/f < 0.9.$$

9. The optical system for imaging pickup of claim 8, wherein the first lens element has a convex object-side surface, the fourth lens element has a concave object-side surface and a convex image-side surface, and the fourth lens element and the fifth lens element are made of plastic.

10. The optical system for imaging pickup of claim 8, wherein R_1 is the curvature radius of the object-side surface of the first lens element, R_2 is the curvature radius of the image-side surface of the first lens element, and the following relation is satisfied:

$$|R_1/R_2| < -2.0.$$

11. The optical system for imaging pickup of claim 8, wherein f_2 is a focal length of the second lens element, f_3 is a focal length of the third lens element, and the following relation is satisfied:

$$|f_2/f_3| < 0.5.$$

12. The optical system for imaging pickup of claim 8, wherein f is the focal length of the optical system for imaging pickup, f_4 is the focal length of the fourth lens element, f_5 is the focal length of the fifth lens element, and the following relation is satisfied:

$$3.0 < f/f_4 + f/f_5 < 5.0.$$

13. The optical system for imaging pickup of claim 3, further comprising an image sensor installed at an image plane for imaging an object, wherein TTL is an axial distance

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between the object-side surface of the first lens element and the image plane, ImgH is half of a diagonal length of an effective photosensitive area of the image sensor, and the following relation is satisfied:

$$TTL/\text{ImgH} < 2.1.$$

14. An optical system for imaging pickup, sequentially arranged from an object side to an image side, comprising:

a first lens element with positive refractive power, having a convex image-side surface;

a second lens element with refractive power, having a concave image-side surface;

a third lens element with refractive power;

a fourth lens element with positive refractive power, having a convex image-side surface, both object-side surface and image-side surface being aspheric, and at least one of the object-side surface and image-side surface having at least one inflection point; and

a fifth lens element with negative refractive power, having a concave image-side surface, both object-side surface and image-side surface being aspheric, and at least one of the object-side surface and the image-side surface having at least one inflection point, wherein R_1 is a curvature radius of the object-side surface of the first lens element, R_2 is a curvature radius of the image-side surface of the first lens element, f is a focal length of the optical system for imaging pickup, f_1 is a focal length of the first lens element, f_5 is a focal length of the fifth lens element, and the following relations are satisfied:

$$|R_1/R_2| < -1.0;$$

$$0.5 < f_1/f < 1.4; \text{ and}$$

$$-1.5 < f_5/f < -0.3.$$

15. The optical system for imaging pickup of claim 14, wherein the second lens element has negative refractive power, and the fourth lens element and the fifth lens element are made of plastic.

16. The optical system for imaging pickup of claim 15, wherein R_2 is the curvature radius of the image-side surface of the first lens element, R_4 is a curvature radius of the image-side surface of the second lens element, and the following relation is satisfied:

$$-2.0 < R_2/R_4 < -0.3.$$

17. The optical system for imaging pickup of claim 15, wherein R_1 is the curvature radius of the object-side surface of the first lens element, R_2 is the curvature radius of the image-side surface of the first lens element, and the following relation is satisfied:

$$|R_1/R_2| < -2.0.$$

18. The optical system for imaging pickup of claim 15, wherein the first lens element has a convex object-side surface; f is the focal length of the optical system for imaging pickup, f_4 is a focal length of the fourth lens element, f_5 is the focal length of the fifth lens element, and the following relation is satisfied:

$$3.0 < f/f_4 + f/f_5 < 5.0.$$

19. An optical system for imaging pickup, sequentially arranged from an object side to an image side, comprising:

a first lens element with positive refractive power, having a convex image-side surface;

a second lens element with negative refractive power;

a third lens element with refractive power;

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a fourth lens element with positive refractive power, having both object-side surface and image-side surface being aspheric, and at least one of the object-side surface and the image-side surface having at least one inflection point; and

a fifth lens element with negative refractive power, having a concave image-side surface, both object-side surface and image-side surface being aspheric, and at least one of the object-side surface and the image-side surface having at least one inflection point, wherein R_1 is a curvature radius of the object-side surface of the first lens element, R_2 is a curvature radius of the image-side surface of the first lens element, f is a focal length of the optical system for imaging pickup, f_1 is a focal length of the first lens element, f_4 is a focal length of the fourth lens element, f_5 is a focal length of the fifth lens element, and the following relations are satisfied:

$$|R_1|/R_2 < -1.0;$$

$$0.5 < f_1/f < 1.4;$$

$$0.35 < f_4/f < 0.67; \text{ and}$$

$$-1.5 < f_5/f < -0.3.$$

20. The optical system for imaging pickup of claim **19**, further comprising a stop, wherein the fourth lens element has a concave object-side surface and a convex image-side sur-

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face; S_D is an axial distance between the stop and the image-side surface of the fifth lens element, T_D is an axial distance between the object-side surface of the first lens element and the image-side surface of the fifth lens element, f_2 is a focal length of the second lens element, f_3 is a focal length of the third lens element, and the following relations are satisfied:

$$0.7 < S_D/T_D < 1.15; \text{ and}$$

$$|f_2/f_3| < 0.5.$$

21. The optical system for imaging pickup of claim **20**, wherein R_2 is the curvature radius of the image-side surface of the first lens element, R_4 is a curvature radius of the image-side surface of the second lens element, and the following relation is satisfied:

$$-2.0 < R_2/R_4 < -0.3.$$

22. The optical system for imaging pickup of claim **19**, further comprising an image sensor, wherein ImgH is half of a diagonal length of an effective photosensitive area of the image sensor, f is the focal length of the optical system for imaging pickup, and the following relation is satisfied:

$$0.6 < \text{ImgH}/f < 1.0.$$

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